
Statistics and Statistical Learning

Technical reference

Probability, standard random variables, sampling distributions, estimation, intervals and tests, with proofs. Each chapter says which site note explains it; the last chapter collects the formulas.

Companion to the notes at <https://lorenzoliuzzo.github.io/notes/>

CONTENTS

1	Probability toolkit	2
1.1	Probability spaces and the axioms	2
1.2	Conditional probability, independence, Bayes	3
1.3	Random variables and distributions	4
1.4	Expectation	4
1.5	Variance, covariance, correlation	4
1.6	Markov and Chebyshev	5
1.7	Limit theorems	5
2	Discrete random variables	6
2.1	Bernoulli and binomial	6
2.2	Geometric	7
2.3	Poisson	7
3	Continuous random variables	8
3.1	Uniform and inverse transform sampling	8
3.2	Exponential	8
3.3	Normal	8
3.4	Gamma and chi-square	9
4	Sampling distributions of statistics	9
4.1	Mean, variance and proportion	9
4.2	Normal-theory distributions	10
4.3	The t and F distributions in more detail	10
4.4	A non-normal example: the sample maximum	11
4.5	Simulation and the bootstrap	11
5	Point estimation	12
5.1	Fisher information and the Cramér–Rao bound	12
5.2	Sufficiency (brief)	13
6	Maximum likelihood	13
6.1	Asymptotic properties	14
6.2	Method of moments	14
7	Interval estimation	14
8	Hypothesis testing	15
8.1	Framework	15
8.2	The t -tests	16
8.3	Power and sample size	16
8.4	Likelihood ratio tests	17
8.5	Multiple testing	17
9	Linear regression	18
9.1	The model and least squares	18

9.2	Normal errors and inference	19
9.3	Sums of squares and the F -test	19
9.4	Leverage, influence and collinearity	20
10	Regularization	21
10.1	Prediction error and the trade-off	21
10.2	Ridge regression	21
10.3	Lasso	22
11	Splines and additive models	23
11.1	Basis expansions and regression splines	23
11.2	Smoothing splines	24
11.3	Additive models and backfitting	25
11.4	MARS	25
12	Robust statistics	26
12.1	Breakdown and the influence function	26
12.2	M-estimators	26
12.3	Robust regression	27
13	Summary of the main formulas	28
14	References	28

1 Probability toolkit

In the notes: Probability Foundations · Random Variables, Expectation and Variance

1.1 Probability spaces and the axioms

A probability space is a triple $(\Omega, \mathcal{F}, P)$: a sample space Ω , a σ -algebra $\mathcal{F}$ of events (a family of subsets of Ω containing Ω and closed under complements and countable unions), and a function $P : \mathcal{F} \rightarrow [0, 1]$ satisfying the axioms below. Events A, B are *disjoint* if $A \cap B = \emptyset$; events $B_1, B_2, \dots$ form a *partition* if they are pairwise disjoint and $\bigcup_i B_i = \Omega$.

DEFINITION Kolmogorov axioms (A1) $P(A) \geq 0$ for every event A . (A2) $P(\Omega) = 1$. (A3) For pairwise disjoint events $A_1, A_2, \dots$, $P(\bigcup_i A_i) = \sum_i P(A_i)$.

THEOREM Consequences of the axioms For events $A, B, A_1, A_2, \dots$: (i) $P(\emptyset) = 0$; (ii) finite additivity: $P(\bigcup_{i=1}^n A_i) = \sum_{i=1}^n P(A_i)$ for disjoint A_i ; (iii) $P(A^c) = 1 - P(A)$; (iv) if $A \subseteq B$ then $P(A) \leq P(B)$; in particular $P(A) \leq 1$; (v) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; (vi) union bound: $P(\bigcup_i A_i) \leq \sum_i P(A_i)$.

Proof. (i) Take $A_i = \emptyset$ for all i in (A3): $P(\emptyset) = \sum_i P(\emptyset)$, which forces $P(\emptyset) = 0$. (ii) Apply (A3) with $A_i = \emptyset$ for $i > n$ and use (i). (iii) $\Omega = A \cup A^c$ is a disjoint union, so $1 = P(A) + P(A^c)$. (iv) $B = A \cup (B \setminus A)$ is disjoint, so $P(B) = P(A) + P(B \setminus A) \geq P(A)$; with $B = \Omega$ this gives $P(A) \leq 1$. (v) Write $A \cup B = A \cup (B \setminus A)$ and $B = (A \cap B) \cup (B \setminus A)$, both disjoint; then $P(B \setminus A) = P(B) - P(A \cap B)$, and substituting gives (v). (vi) Set $B_1 = A_1$ and $B_i = A_i \setminus \bigcup_{j < i} A_j$. The B_i are disjoint, $\bigcup_i B_i = \bigcup_i A_i$ and $B_i \subseteq A_i$, so by (A3) and (iv) $P(\bigcup_i A_i) = \sum_i P(B_i) \leq \sum_i P(A_i)$. $\square$

The general inclusion-exclusion formula is $P(\bigcup_{i=1}^n A_i) = \sum_i P(A_i) - \sum_{i < j} P(A_i \cap A_j) + \sum_{i < j < k} P(A_i \cap A_j \cap A_k) - \dots$. Two further facts follow from (A3) by taking limits: if A_n increases to A then $P(A_n) \rightarrow P(A)$, and likewise for decreasing sequences (continuity of P). When Ω is finite and outcomes are equally likely, $P(A) = |A| / |\Omega|$.

1.2 Conditional probability, independence, Bayes

DEFINITION For $P(B) > 0$, the conditional probability of A given B is $P(A | B) = P(A \cap B) / P(B)$. Then $P(\cdot | B)$ is itself a probability: it satisfies (A1)–(A3) on $\mathcal{F}$, so all the consequences above hold conditionally.

Rearranging gives the multiplication rule $P(A \cap B) = P(A | B) P(B)$ and, by induction, the chain rule

$$P(A_1 \cap \dots \cap A_n) = P(A_1) P(A_2 | A_1) \dots P(A_n | A_1 \cap \dots \cap A_{n-1}). \quad (1)$$

DEFINITION Independence A and B are independent if $P(A \cap B) = P(A) P(B)$ (equivalently $P(A | B) = P(A)$ when $P(B) > 0$). Events $A_1, \dots, A_n$ are *mutually* independent if $P(\bigcap_{i \in S} A_i) = \prod_{i \in S} P(A_i)$ for every subset S of indices; pairwise independence alone does not imply this. A and B are conditionally independent given C if $P(A \cap B | C) = P(A | C) P(B | C)$.

Remark. Disjoint events with positive probability are never independent: $P(A \cap B) = 0 \neq P(A) P(B)$.

THEOREM Law of total probability and Bayes' theorem Let $B_1, B_2, \dots$ be a partition of Ω with $P(B_i) > 0$. For any event A ,

$$P(A) = \sum_i P(A | B_i) P(B_i), \quad (2)$$

and, if $P(A) > 0$,

$$P(B_j | A) = \frac{P(A | B_j) P(B_j)}{\sum_i P(A | B_i) P(B_i)}. \quad (3)$$

Proof. The sets $A \cap B_i$ are disjoint with union A , so by (A3) and the multiplication rule $P(A) = \sum_i P(A \cap B_i) = \sum_i P(A | B_i) P(B_i)$. For Bayes, $P(B_j | A) = P(A \cap B_j) / P(A) = P(A | B_j) P(B_j) / P(A)$, and the denominator is Equation 2. $\square$

In Bayesian language $P(B_j)$ is the prior, $P(A | B_j)$ the likelihood, $P(A)$ the evidence (marginal likelihood) and $P(B_j | A)$ the posterior. The odds form is $P(B | A) / P(B^c | A) = [P(A | B) / P(A | B^c)] \cdot [P(B) / P(B^c)]$: posterior odds equal likelihood ratio times prior odds.

EXAMPLE Diagnostic test Prevalence $P(D) = 0.01$, sensitivity $P(+ | D) = 0.95$, false-positive rate $P(+ | D^c) = 0.10$. Then

$$P(D | +) = \frac{0.95 \cdot 0.01}{0.95 \cdot 0.01 + 0.10 \cdot 0.99} = \frac{0.0095}{0.1085} \approx 0.088. \quad (4)$$

The posterior is below 9% because the healthy majority produces more false positives than the sick minority produces true ones: the base rate matters.

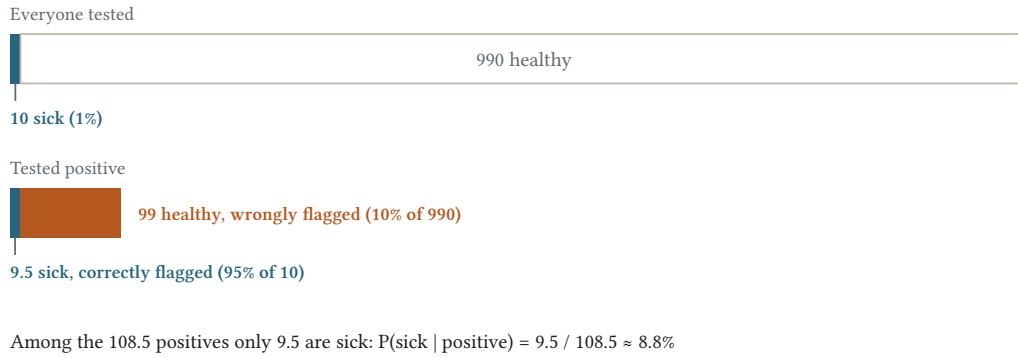


Figure 1: The same test counted in people. The sick are a thin sliver of the population, so the 10% of healthy people who test positive outnumber the true cases about ten to one.

1.3 Random variables and distributions

A random variable is a measurable function $X : \Omega \rightarrow \mathbb{R}$. Its distribution function is $F(x) = P(X \leq x)$.

LEMMA Properties of the distribution function F is non-decreasing and right-continuous, with $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$. For $a < b$, $P(a < X \leq b) = F(b) - F(a)$.

Proof. Monotonicity is (iv) applied to $\{X \leq x\} \subseteq \{X \leq y\}$. Right-continuity and the limits follow from continuity of P along monotone sequences of events. Finally $\{X \leq b\} = \{X \leq a\} \cup \{a < X \leq b\}$ is disjoint. $\square$

X is discrete with mass function $p(x) = P(X = x)$, $\sum_x p(x) = 1$, or continuous with density $f = F'$, $\int f = 1$ and $P(a \leq X \leq b) = \int_a^b f$; in the continuous case $P(X = x) = 0$ for every x . The quantile function is $F^{-1}(p) = \inf\{x : F(x) \geq p\}$ and the median is $F^{-1}(1/2)$. For a pair (X, Y) with joint density $f_{X,Y}$, the marginal is $f_{X(x)} = \int f_{X,Y}(x, y) dy$, the conditional density is $f_{X|Y}(x | y) = f_{X,Y}(x, y) / f_Y(y)$, and X, Y are independent iff $f_{X,Y} = f_X f_Y$.

1.4 Expectation

DEFINITION $E[X] = \sum_x x p(x)$ (discrete) or $\int x f(x) dx$ (continuous), when the sum or integral converges absolutely. For a function g (law of the unconscious statistician), $E[g(X)] = \sum_x g(x) p(x)$ or $\int g(x) f(x) dx$.

THEOREM Properties of expectation (i) Linearity: $E[aX + bY + c] = a E[X] + b E[Y] + c$ for any X, Y , dependent or not. (ii) Monotonicity: $X \leq Y$ implies $E[X] \leq E[Y]$. (iii) If X, Y are independent, $E[XY] = E[X] E[Y]$. (iv) $E[\mathbb{1}_A] = P(A)$. (v) Jensen: for convex g , $g(E[X]) \leq E[g(X)]$.

Proof. (i) and (ii) follow from linearity and monotonicity of sums and integrals, using the joint density for X and Y together. (iii) By independence $f_{X,Y} = f_X f_Y$, so $E[XY] = \int \int xy f_{X(x)} f_{Y(y)} dx dy = E[X] E[Y]$. (iv) $\mathbb{1}_A$ takes the value 1 with probability $P(A)$ and 0 otherwise. (v) A convex g lies above its tangent line at $m = E[X]$: $g(x) \geq g(m) + c(x - m)$ for some slope c ; take expectations. $\square$

1.5 Variance, covariance, correlation

DEFINITION $\text{Var}(X) = E[(X - \mu)^2]$ with $\mu = E[X]$; the standard deviation is $\sigma = \sqrt{\text{Var}(X)}$. $\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$ and, when $\sigma_X, \sigma_Y > 0$, the correlation is $\rho_{XY} = \text{Cov}(X, Y) / (\sigma_X \sigma_Y)$. Moments: $E[X^k]$; central moments $E[(X - \mu)^k]$; skewness $E[(X - \mu)^3] / \sigma^3$ and kurtosis $E[(X - \mu)^4] / \sigma^4$ (equal to 3 for the normal).

THEOREM Properties of variance and covariance (i) $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E} X)^2 \geq 0$, with equality iff X is almost surely constant. (ii) $\text{Var}(aX + b) = a^2 \text{Var}(X)$. (iii) $\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X] \mathbb{E}[Y]$, $\text{Cov}(X, X) = \text{Var}(X)$, and Cov is bilinear. (iv) $\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y)$. (v) Independent X, Y have $\text{Cov}(X, Y) = 0$; the converse is false. (vi) $|\rho_{XY}| \leq 1$, with equality iff $Y = aX + b$ almost surely.

Proof. (i) Expand $(X - \mu)^2 = X^2 - 2\mu X + \mu^2$ and take expectations; nonnegativity holds because the integrand is ≥ 0 , and $\mathbb{E}[(X - \mu)^2] = 0$ forces $X = \mu$ almost surely. (ii) $aX + b - \mathbb{E}[aX + b] = a(X - \mu)$. (iii) Expand the product and use linearity; bilinearity is linearity in each argument. (iv) is (ii) and (iii) combined. (v) follows from (iii) and property (iii) of expectation. For the failure of the converse, let $X \sim \mathcal{N}(0, 1)$ and $Y = X^2$: $\text{Cov}(X, Y) = \mathbb{E}[X^3] - \mathbb{E}[X] \mathbb{E}[X^2] = 0$ since odd moments of a symmetric law vanish, yet Y is a function of X . (vi) For every real t , $0 \leq \text{Var}(Y - tX) = \text{Var}(Y) - 2t \text{Cov}(X, Y) + t^2 \text{Var}(X)$. A quadratic in t that never goes negative has non-positive discriminant, $4\text{Cov}(X, Y)^2 - 4 \text{Var}(X) \text{Var}(Y) \leq 0$, which is $|\rho| \leq 1$. Equality means the discriminant vanishes, so $\text{Var}(Y - tX) = 0$ for some t and $Y - tX$ is constant. $\square$

Property (iv) generalizes to $\text{Var}(\sum_i X_i) = \sum_i \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j)$, and so, for uncorrelated (hence independent) X_i with common variance σ^2 ,

$$\text{Var}\left(\sum_i X_i\right) = n\sigma^2, \quad \text{Var}(\bar{X}_n) = \sigma^2/n. \quad (5)$$

THEOREM Law of total expectation and total variance If $\mathbb{E}[X^2] < \infty$, then $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X | Y]]$ and $\text{Var}(X) = \mathbb{E}[\text{Var}(X | Y)] + \text{Var}(\mathbb{E}[X | Y])$. $\square$

Proof. The first identity is the law of total probability for expectations: $\mathbb{E}[X] = \int \mathbb{E}[X | Y = y] f_{Y(y)} dy$. For the second, write $X - \mathbb{E} X = (X - \mathbb{E}[X | Y]) + (\mathbb{E}[X | Y] - \mathbb{E} X)$ and square. The cross term has expectation zero: conditionally on Y the second factor is fixed and $\mathbb{E}[X - \mathbb{E}[X | Y] | Y] = 0$. The first square has expectation $\mathbb{E}[\text{Var}(X | Y)]$ and the second $\text{Var}(\mathbb{E}[X | Y])$. $\square$

1.6 Markov and Chebyshev

LEMMA Markov and Chebyshev If $X \geq 0$ and $a > 0$ then $\mathbb{P}(X \geq a) \leq \mathbb{E}[X]/a$. For any X with mean μ and variance σ^2 and any $k > 0$, $\mathbb{P}(|X - \mu| \geq k) \leq \sigma^2/k^2$.

Proof. $a \mathbb{1}\{X \geq a\} \leq X$ pointwise, take expectations. Chebyshev is Markov applied to $(X - \mu)^2$ and $a = k^2$. $\square$

The standard discrete and continuous distributions, with their means, variances and uses, are in the chapters on discrete and continuous random variables below.

1.7 Limit theorems

Write $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$.

THEOREM Weak law of large numbers If X_i are i.i.d. with $\mathbb{E} X_1 = \mu$ and $\text{Var}(X_1) = \sigma^2 < \infty$, then $\bar{X}_n \xrightarrow{P} \mu$, i.e. $\mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \rightarrow 0$ for every $\varepsilon > 0$.

Proof. By Equation 5, $\text{Var}(\bar{X}_n) = \sigma^2/n$. Chebyshev gives $\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \sigma^2/(n\varepsilon^2) \rightarrow 0$. $\square$

THEOREM Central limit theorem If X_i are i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$, then

$$\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow{d} \mathcal{N}(0, 1). \quad (7)$$

The proof goes through characteristic functions and is not reproduced here. The Berry–Esseen theorem quantifies the rate: the Kolmogorov distance to Φ is at most $C E|X_1 - \mu|^3 / (\sigma^3 \sqrt{n})$. This is why the sample size needed depends on skewness.

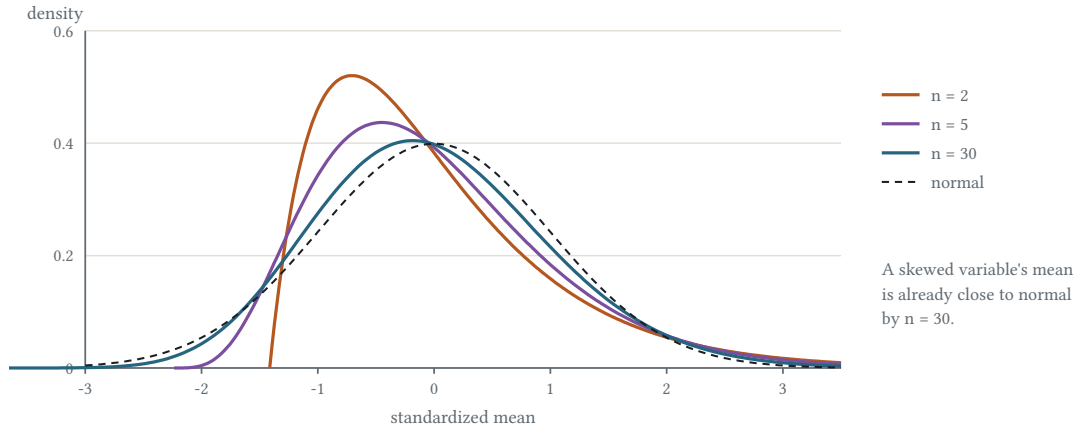


Figure 2: Standardized mean of n exponential variables against the standard normal density. At $n = 2$ the skew is obvious, by $n = 30$ the two curves are hard to tell apart.

RULE OF THUMB $n \geq 30$ is a guide, not a theorem: the Berry–Esseen bound grows with skewness, and for heavy-tailed data it takes far more.

THEOREM Slutsky and delta method If $A_n \xrightarrow{d} A$ and $B_n \xrightarrow{P} b$ (constant), then $A_n + B_n \xrightarrow{d} A + b$ and $A_n B_n \xrightarrow{d} bA$. If moreover $\sqrt{n}(T_n - \theta) \xrightarrow{d} \mathcal{N}(0, \tau^2)$ and g is differentiable at θ with $g'(\theta) \neq 0$, then

$$\sqrt{n}(g(T_n) - g(\theta)) \xrightarrow{d} \mathcal{N}(0, g'(\theta)^2 \tau^2). \quad (8)$$

Slutsky is what licenses replacing an unknown σ by a consistent estimate in a pivot, which is how the Wald and large-sample t intervals are justified.

2 Discrete random variables

In the notes: Common Random Variables

A discrete random variable has mass function $p(x) = P(X = x)$ on a countable support, with $\sum_x p(x) = 1$, $E[X] = \sum_x x p(x)$ and $\text{Var}(X) = E[X^2] - (E[X])^2$. Probability background is in the Probability toolkit chapter.

2.1 Bernoulli and binomial

DEFINITION $X \sim \text{Bernoulli}(p)$ if $P(X = 1) = p$ and $P(X = 0) = 1 - p$. $Y \sim \text{Binomial}(n, p)$ is the sum of n independent $\text{Bernoulli}(p)$ variables, with mass function $P(Y = k) = \binom{n}{k} p^k (1 - p)^{n-k}$, $k = 0, \dots, n$.

THEOREM Mean and variance If $X \sim \text{Bernoulli}(p)$ then $E[X] = p$ and $\text{Var}(X) = p(1 - p)$. If $Y \sim \text{Binomial}(n, p)$ then $E[Y] = np$ and $\text{Var}(Y) = np(1 - p)$.

Proof. $E[X] = 1 \cdot p + 0 \cdot (1 - p) = p$, and since $X^2 = X$, $E[X^2] = p$ and $\text{Var}(X) = p - p^2$. For $Y = \sum_{i=1}^n X_i$ with independent X_i , linearity gives $E[Y] = np$ and independence gives $\text{Var}(Y) = \sum_i \text{Var}(X_i) = np(1 - p)$. $\square$

The binomial mass function counts the $\binom{n}{k}$ arrangements of k successes among n trials, each with probability $p^k(1 - p)^{n-k}$. For large n it is approximately $\mathcal{N}(np, np(1 - p))$ by the central limit theorem; a common rule is $np(1 - p) \geq 10$.

2.2 Geometric

DEFINITION $X \sim \text{Geometric}(p)$ counts the trials up to and including the first success: $P(X = k) = (1 - p)^{k-1}p$, $k = 1, 2, \dots$

THEOREM Mean and memorylessness $E[X] = 1/p$ and $\text{Var}(X) = (1 - p)/p^2$. Moreover $P(X > m + k \mid X > m) = P(X > k)$ for all integers $m, k \geq 0$.

Proof. With $q = 1 - p$, $E[X] = p \sum_{k \geq 1} kq^{k-1} = p/(1 - q)^2 = 1/p$, using $\sum_{k \geq 1} kq^{k-1} = \frac{d}{dq} \sum_{k \geq 0} q^k = (1 - q)^{-2}$. A second differentiation gives $E[X(X - 1)]$ and hence the variance. For the tail, $P(X > k) = q^k$, so $P(X > m + k \mid X > m) = q^{m+k}/q^m = q^k$. $\square$

2.3 Poisson

DEFINITION $X \sim \text{Poisson}(\lambda)$ has $P(X = k) = e^{-\lambda} \lambda^k / k!$, $k = 0, 1, 2, \dots$

THEOREM Moments, sums and the binomial limit (i) $E[X] = \text{Var}(X) = \lambda$. (ii) If $X \sim \text{Poisson}(\lambda)$ and $Y \sim \text{Poisson}(\mu)$ are independent then $X + Y \sim \text{Poisson}(\lambda + \mu)$. (iii) If $X_n \sim \text{Binomial}(n, \lambda/n)$ then $P(X_n = k) \rightarrow e^{-\lambda} \lambda^k / k!$ for every fixed k .

Proof. (i) $E[X] = \sum_{k \geq 1} k e^{-\lambda} \lambda^k / k! = \lambda e^{-\lambda} \sum_{k \geq 1} \lambda^{k-1} / (k-1)! = \lambda$, and similarly $E[X(X - 1)] = \lambda^2$, so $\text{Var}(X) = \lambda^2 + \lambda - \lambda^2 = \lambda$. (ii) $P(X + Y = m) = \sum_{j=0}^m e^{-\lambda} \lambda^j / j! \cdot e^{-\mu} \mu^{m-j} / (m - j)! = e^{-(\lambda + \mu)} (\lambda + \mu)^m / m!$ by the binomial theorem. (iii) $P(X_n = k) = [n! / ((n - k)! n^k)] \cdot (\lambda^k / k!) (1 - \lambda/n)^{n-k}$. As $n \rightarrow \infty$ the first factor tends to 1, $(1 - \lambda/n)^n \rightarrow e^{-\lambda}$ and $(1 - \lambda/n)^{-k} \rightarrow 1$. $\square$

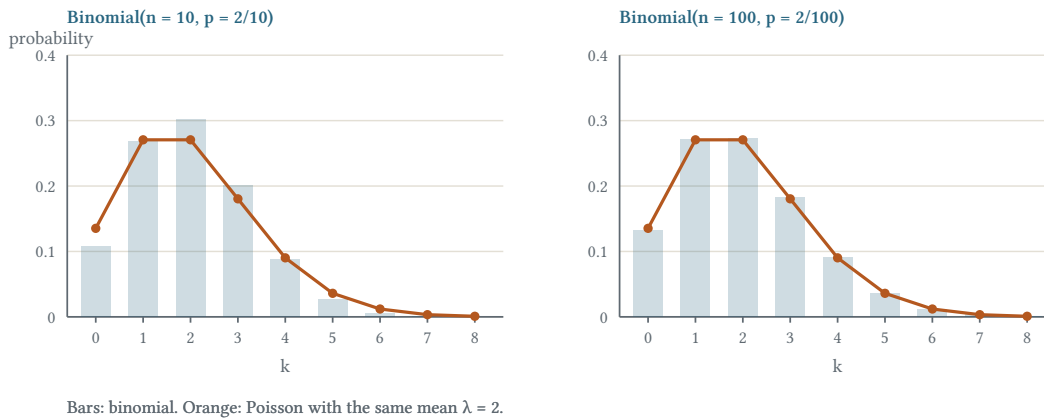


Figure 3: Binomial bars with the Poisson of the same mean $\lambda = 2$ on top. With $n = 10$ they differ visibly, with $n = 100$ they nearly coincide.

3 Continuous random variables

In the notes: Common Random Variables

A continuous random variable has a density f with $P(a \leq X \leq b) = \int_a^b f$, mean $E[X] = \int x f(x) dx$ and distribution function $F(x) = \int_{-\infty}^x f$.

3.1 Uniform and inverse transform sampling

$X \sim \text{Uniform}(a, b)$ has $f(x) = 1/(b-a)$ on $[a, b]$, $E[X] = (a+b)/2$ and $\text{Var}(X) = (b-a)^2/12$ (compute $E[X^2] = (a^2 + ab + b^2)/3$).

THEOREM Inverse transform Let F be a distribution function, $F^{-1}(u) = \inf\{x : F(x) \geq u\}$, and $U \sim \text{Uniform}(0, 1)$. Then $F^{-1}(U)$ has distribution function F . Conversely, if X has a continuous distribution function F , then $F(X) \sim \text{Uniform}(0, 1)$.

Proof. The generalized inverse satisfies $F^{-1}(u) \leq x$ iff $u \leq F(x)$, so $P(F^{-1}(U) \leq x) = P(U \leq F(x)) = F(x)$. For the converse, with F continuous and strictly increasing on the support, $P(F(X) \leq u) = P(X \leq F^{-1}(u)) = F(F^{-1}(u)) = u$. $\square$

The converse is the probability integral transform used for the uniformity of p -values; the direct statement is how simulation software turns uniform random numbers into draws from any distribution with a computable F^{-1} .

3.2 Exponential

DEFINITION $X \sim \text{Exponential}(\lambda)$ has $f(x) = \lambda e^{-\lambda x}$, $x \geq 0$, so that $P(X > x) = e^{-\lambda x}$.

THEOREM Moments and memorylessness $E[X] = 1/\lambda$, $\text{Var}(X) = 1/\lambda^2$, and $P(X > s+t \mid X > s) = P(X > t)$.

Proof. Integrating by parts, $E[X] = \int_0^\infty x \lambda e^{-\lambda x} dx = 1/\lambda$ and $E[X^2] = 2/\lambda^2$, so $\text{Var}(X) = 1/\lambda^2$. For the tail, $P(X > s+t \mid X > s) = e^{-\lambda(s+t)}/e^{-\lambda s} = e^{-\lambda t}$. $\square$

The exponential is the only continuous distribution on $[0, \infty)$ with this property. If events arrive as a Poisson process of rate λ (the count in an interval of length t is $\text{Poisson}(\lambda t)$), the waiting time to the first event, and between successive events, is $\text{Exponential}(\lambda)$: $P(T > t) = P(\text{no event in } [0, t]) = e^{-\lambda t}$.

3.3 Normal

DEFINITION $X \sim \mathcal{N}(\mu, \sigma^2)$ has density $(2\pi\sigma^2)^{-1/2} \exp(-(x-\mu)^2/(2\sigma^2))$. The standard normal $\mathcal{N}(0, 1)$ has distribution function Φ and quantiles $z_p = \Phi^{-1}(p)$.

THEOREM Standardization and closure (i) $E[X] = \mu$ and $\text{Var}(X) = \sigma^2$. (ii) $Z = (X - \mu)/\sigma \sim \mathcal{N}(0, 1)$, so $P(X \leq x) = \Phi((x - \mu)/\sigma)$. (iii) If $X_1, \dots, X_n$ are independent, $X_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$, then $\sum_i a_i X_i + b \sim \mathcal{N}(\sum_i a_i \mu_i + b, \sum_i a_i^2 \sigma_i^2)$.

Proof. (ii) The substitution $y = (u - \mu)/\sigma$ in $\int_{-\infty}^x f(u) du$ gives $\int_{-\infty}^{(x-\mu)/\sigma} \varphi(y) dy$. (i) follows from (ii) since $E[Z] = 0$ (odd integrand) and $E[Z^2] = 1$ (integration by parts). (iii) The mean and variance follow from linearity and independence; that the sum is again normal follows from the moment generating function $E[e^{tX}] = e^{\mu t + \sigma^2 t^2/2}$, which multiplies for independent summands. $\square$

Quantiles that recur: $z_{0.90} \approx 1.282$, $z_{0.95} \approx 1.645$, $z_{0.975} \approx 1.960$, $z_{0.995} \approx 2.576$, and $P(|Z| \leq 1, 2, 3) \approx 0.683, 0.954, 0.997$.

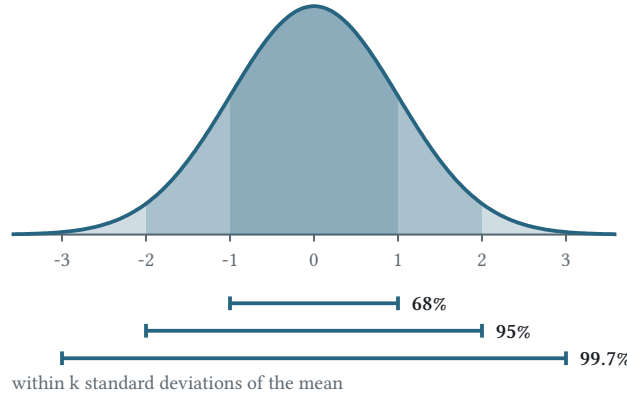


Figure 4: The 68, 95 and 99.7 percent rule.

3.4 Gamma and chi-square

DEFINITION $X \sim \text{Gamma}(\alpha, \beta)$ (shape $\alpha > 0$, rate $\beta > 0$) has density $\beta^\alpha x^{\alpha-1} e^{-\beta x} / \Gamma(\alpha)$ on $x > 0$, with $E[X] = \alpha/\beta$ and $\text{Var}(X) = \alpha/\beta^2$. The sum of k independent $\text{Exponential}(\lambda)$ variables is $\text{Gamma}(k, \lambda)$. The chi-square distribution is $\chi_k^2 = \text{Gamma}(k/2, 1/2)$, with density $x^{k/2-1} e^{-x/2} / (2^{k/2} \Gamma(k/2))$.

THEOREM Chi-square as a sum of squares If $Z_1, \dots, Z_k$ are independent $\mathcal{N}(0, 1)$ then $\sum_i Z_i^2 \sim \chi_k^2$, with mean k and variance $2k$.

Proof. The density of Z_1^2 follows by the change of variable $y = z^2$ and equals the χ_1^2 density; independent Gamma variables with a common rate add their shapes, giving χ_k^2 . For the moments, $E[Z^2] = 1$ and $\text{Var}(Z^2) = E[Z^4] - 1 = 3 - 1 = 2$, and the Z_i^2 are independent, so the mean is k and the variance $2k$. $\square$

4 Sampling distributions of statistics

In the notes: Sampling Distributions

A statistic $T = T(X_1, \dots, X_n)$ is a random variable. Its sampling distribution is the law of T when the sample is drawn from the assumed model, and its standard error is $\text{se}(T) = \sqrt{\text{Var}(T)}$.

4.1 Mean, variance and proportion

THEOREM Mean and variance of $\bar{X}$ For i.i.d. X_i with mean μ and variance σ^2 , $E[\bar{X}] = \mu$ and $\text{Var}(\bar{X}) = \sigma^2/n$. For two independent samples, $\bar{X}_1 - \bar{X}_2$ has mean $\mu_1 - \mu_2$ and variance $\sigma_1^2/n_1 + \sigma_2^2/n_2$.

Proof. Linearity gives $E[\bar{X}] = n^{-1} \sum_i E[X_i] = \mu$; independence gives $\text{Var}(\bar{X}) = n^{-2} \sum_i \text{Var}(X_i) = \sigma^2/n$. For the difference, $\text{Var}(\bar{X}_1 - \bar{X}_2) = \text{Var}(\bar{X}_1) + \text{Var}(\bar{X}_2)$ by independence. $\square$

If the data are normal, $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$ exactly; in general the central limit theorem gives $\sqrt{n}(\bar{X} - \mu)/\sigma \xrightarrow{d} \mathcal{N}(0, 1)$.

THEOREM Sample proportion If $Y \sim \text{Binomial}(n, p)$ and $\hat{p} = Y/n$, then $E[\hat{p}] = p$, $\text{Var}(\hat{p}) = p(1-p)/n$, and $\sqrt{n}(\hat{p} - p)/\sqrt{p(1-p)} \xrightarrow{d} \mathcal{N}(0, 1)$.

Proof. The moments follow from those of the binomial and the scaling rule $\text{Var}(Y/n) = \text{Var}(Y)/n^2$. The limit is the central limit theorem applied to $Y = \sum_i X_i$ with Bernoulli summands (the de Moivre–Laplace theorem). $\square$

EXAMPLE Classifier accuracy A classifier with true accuracy $p = 0.8$ evaluated on $n = 100$ independent test examples has $\text{se}(\hat{p}) = \sqrt{0.8 \cdot 0.2/100} = 0.04$, so the observed accuracy lies in 0.8 ± 0.08 with probability about 95%.

4.2 Normal-theory distributions

DEFINITION Gaussian family $X \sim \mathcal{N}(\mu, \sigma^2)$ has density $(2\pi\sigma^2)^{-1/2} \exp(-(x - \mu)^2/(2\sigma^2))$. Let $Z_1, \dots, Z_k \sim \mathcal{N}(0, 1)$ be independent. Then

- $\sum_i Z_i^2 \sim \chi_k^2$ (chi-square, k degrees of freedom), with mean k and variance $2k$;
- if $Z \sim \mathcal{N}(0, 1)$ and $V \sim \chi_k^2$ are independent, $Z/\sqrt{V/k} \sim t_k$ (Student);
- if $V_1 \sim \chi_a^2$ and $V_2 \sim \chi_b^2$ are independent, $(V_1/a)/(V_2/b) \sim F_{a,b}$.

Facts used below: a linear combination of jointly Gaussian variables is Gaussian; for jointly Gaussian variables, uncorrelated implies independent; $t_k \rightarrow \mathcal{N}(0, 1)$ as $k \rightarrow \infty$; $t_k^2 \sim F_{1,k}$.

Let $X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma^2)$ be i.i.d., with $\bar{X} = \bar{X}_n$ and $S^2 = (n-1)^{-1} \sum_i (X_i - \bar{X})^2$.

THEOREM Distribution of $\bar{X}$ and S^2 (i) $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$. (ii) $\bar{X}$ and S^2 are independent. (iii) $(n-1)S^2/\sigma^2 \sim \chi_{n-1}^2$. (iv) Consequently

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}. \quad (9)$$

Proof. (i) is closure of the Gaussian family under linear maps. For (ii)–(iii), let $Z_i = (X_i - \mu)/\sigma$ and Q an orthogonal $n \times n$ matrix whose first row is $(1/\sqrt{n}, \dots, 1/\sqrt{n})$. Then $Y = QZ$ has $Y_1 = \sqrt{n}\bar{Z}$ and, since Q is orthogonal, $Y \sim \mathcal{N}(0, I_n)$: the Y_i are independent standard normals. Orthogonality preserves the norm: $\sum_i Z_i^2 = \sum_i Y_i^2$, and $\sum_i Z_i^2 = \sum_i (Z_i - \bar{Z})^2 + n\bar{Z}^2$, hence

$$\sum_i (Z_i - \bar{Z})^2 = \sum_{i=2}^n Y_i^2 \sim \chi_{n-1}^2, \quad (10)$$

a function of $(Y_2, \dots, Y_n)$ only, and so independent of Y_1 and thus of $\bar{Z}$. Since $\sum_i (Z_i - \bar{Z})^2 = (n-1)S^2/\sigma^2$ and $\bar{X} = \mu + \sigma\bar{Z}$, (ii) and (iii) follow. For (iv), write $T = [(\bar{X} - \mu)/(\sigma/\sqrt{n})] / \sqrt{((n-1)S^2/\sigma^2)/(n-1)}$, a standard normal over the root of an independent $\chi_{n-1}^2/(n-1)$. $\square$

THEOREM Unbiasedness of S^2 For any i.i.d. sample with finite variance σ^2 , $E[S^2] = \sigma^2$.

Proof. $\sum_i (X_i - \bar{X})^2 = \sum_i (X_i - \mu)^2 - n(\bar{X} - \mu)^2$. Taking expectations, $E[\sum_i (X_i - \bar{X})^2] = n\sigma^2 - n(\sigma^2/n) = (n-1)\sigma^2$. $\square$

4.3 The t and F distributions in more detail

The t_k distribution has density $\Gamma((k+1)/2)/(\sqrt{k\pi}\Gamma(k/2)) \cdot (1 + t^2/k)^{-(k+1)/2}$. It is symmetric about 0, has mean 0 for $k > 1$ and variance $k/(k-2)$ for $k > 2$, and has heavier tails than $\mathcal{N}(0, 1)$, to which it converges

as $k \rightarrow \infty$. Selected 97.5% quantiles are $t_{5,0.975} = 2.571$, $t_{9,0.975} = 2.262$, $t_{29,0.975} = 2.045$, against $z_{0.975} = 1.960$.

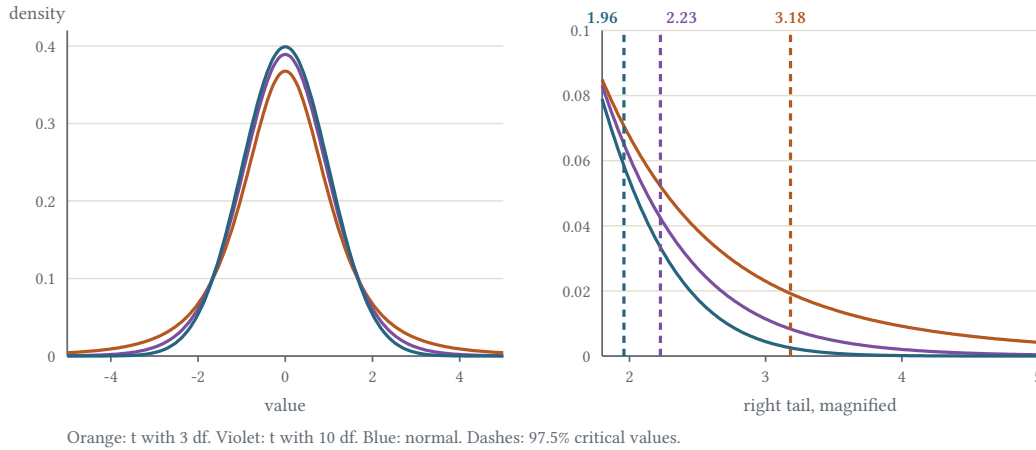


Figure 5: Left, the whole density: the normal and two t curves look alike. Right, the right tail magnified: the t puts more mass out there, so its critical value is larger and its intervals are wider.

LEMMA Relation between t and F If $T \sim t_k$ then $T^2 \sim F_{1,k}$.

Proof. Write $T = Z/\sqrt{V/k}$ with $Z \sim \mathcal{N}(0,1)$ independent of $V \sim \chi_k^2$. Then $T^2 = Z^2/(V/k) = (Z^2/1)/(V/k)$ with $Z^2 \sim \chi_1^2$, the ratio defining $F_{1,k}$. $\square$

The $F_{a,b}$ distribution has mean $b/(b-2)$ for $b > 2$ and is the null distribution of the ratio of two independent normal-sample variance estimates, S_1^2/S_2^2 scaled by σ_2^2/σ_1^2 , with $a = n_1 - 1$ and $b = n_2 - 1$; it is also the null distribution of the overall test in analysis of variance and linear regression.

4.4 A non-normal example: the sample maximum

THEOREM Maximum of uniform samples Let $X_1, \dots, X_n$ be i.i.d. $\text{Uniform}(0, \theta)$ and $M = \max_i X_i$. Then $P(M \leq m) = (m/\theta)^n$ for $0 \leq m \leq \theta$, with density nm^{n-1}/θ^n , $E[M] = n\theta/(n+1)$, and $(n+1)M/n$ is unbiased for θ .

Proof. $M \leq m$ iff every $X_i \leq m$, so by independence $P(M \leq m) = (m/\theta)^n$. Differentiating gives the density, and $E[M] = \int_0^\theta m \cdot nm^{n-1}/\theta^n dm = n\theta/(n+1)$. $\square$

The sampling distribution of M is concentrated against the boundary θ and is far from normal, so normal-theory intervals do not apply: sampling distributions must be derived for the statistic actually used.

4.5 Simulation and the bootstrap

Any sampling distribution can be approximated by simulation: draw B samples from the assumed model, compute T on each, and use the empirical distribution of $T_1, \dots, T_B$; by the law of large numbers its mean and standard deviation converge to $E[T]$ and $\text{se}(T)$ as $B \rightarrow \infty$.

When the model is unknown, the bootstrap replaces the unknown distribution F by the empirical distribution $\hat{F}_n$, which puts mass $1/n$ on each observation. Resample n observations with replacement, compute T_b^* , repeat for $b = 1, \dots, B$, and estimate $\text{se}(T)$ by the standard deviation of the T_b^* . It is consistent for smooth statistics such as the mean; it fails for extreme statistics such as the maximum in the previous example.

5 Point estimation

In the notes: Point Estimation and Maximum Likelihood

This part uses the probability toolkit (axioms, expectation and variance rules, limit theorems, Slutsky) and the sampling distributions of the mean and variance, the χ^2 , t and F laws, from the chapters above. They are quoted without proof.

Let $X_1, \dots, X_n \sim f(\cdot; \theta)$, $\theta \in \Theta \subset \mathbb{R}$, and $\hat{\theta} = T(X_1, \dots, X_n)$.

DEFINITION $\text{bias}(\hat{\theta}) = E_\theta[\hat{\theta}] - \theta$. The estimator is unbiased if this vanishes for all θ . It is consistent if $\hat{\theta}_n \xrightarrow{P} \theta$. Its mean squared error is $\text{MSE}(\hat{\theta}) = E_\theta[(\hat{\theta} - \theta)^2]$.

THEOREM Bias–variance decomposition

$$\text{MSE}(\hat{\theta}) = \text{Var}_\theta(\hat{\theta}) + \text{bias}(\hat{\theta})^2. \quad (11)$$

Proof. Add and subtract $E\hat{\theta}$: $(\hat{\theta} - \theta)^2 = (\hat{\theta} - E\hat{\theta})^2 + 2(\hat{\theta} - E\hat{\theta})(E\hat{\theta} - \theta) + (E\hat{\theta} - \theta)^2$. The cross term has zero expectation because $E\hat{\theta} - \theta$ is a constant and $E[\hat{\theta} - E\hat{\theta}] = 0$. $\square$

EXAMPLE A biased estimator with smaller MSE For $X_i \sim \mathcal{N}(\mu, \sigma^2)$, the estimator $\hat{\sigma}_c^2 = c \sum_i (X_i - \bar{X})^2$ has, using $\sum_i (X_i - \bar{X})^2 \sim \sigma^2 \chi_{n-1}^2$ (mean $(n-1)\sigma^2$, variance $2(n-1)\sigma^4$),

$$\text{MSE} = \sigma^4[2c^2(n-1) + ((n-1)c - 1)^2]. \quad (12)$$

This is minimized at $c = 1/(n+1)$, not at the unbiased choice $c = 1/(n-1)$. The unbiased estimator is therefore not MSE-optimal, a first instance of the shrinkage idea behind Ridge and Lasso.

5.1 Fisher information and the Cramér–Rao bound

Assume the regularity conditions: the support of $f(\cdot; \theta)$ does not depend on θ , $\log f$ is twice differentiable in θ , and differentiation under the integral sign is allowed.

DEFINITION Score and Fisher information The score of one observation is $s(x; \theta) = \partial_\theta \log f(x; \theta)$, and

$$I(\theta) = E_\theta[s(X; \theta)^2] = \text{Var}_\theta(s(X; \theta)). \quad (13)$$

LEMMA Under the regularity conditions, $E_\theta[s(X; \theta)] = 0$ and $I(\theta) = -E_\theta[\partial_\theta^2 \log f(X; \theta)]$. For n independent observations the information is $nI(\theta)$.

Proof. $E[s] = \int (\partial_\theta f/f) f dx = \partial_\theta \int f dx = \partial_\theta 1 = 0$. Differentiating $\int s f dx = 0$ once more gives $\int (\partial_\theta s) f dx + \int s^2 f dx = 0$, and $\partial_\theta s = \partial_\theta^2 \log f$. Additivity follows because the log-likelihood of independent data is the sum of the individual ones, so the scores add and, being independent with mean zero, their variances add. $\square$

THEOREM Cramér–Rao lower bound If $\hat{\theta}$ is unbiased with finite variance, then $\text{Var}_\theta(\hat{\theta}) \geq 1/(nI(\theta))$.

Proof. Let $U = \sum_i s(X_i; \theta)$ be the total score, so $\mathbb{E} U = 0$ and $\text{Var}(U) = nI(\theta)$. Unbiasedness reads $\int \hat{\theta}(x) f_{n(x; \theta)} dx = \theta$ (f_n the joint density). Differentiating in θ gives $\int \hat{\theta}(x) U(x) f_n dx = 1$, i.e. $\text{Cov}(\hat{\theta}, U) = \mathbb{E}[\hat{\theta} U] = 1$ since $\mathbb{E} U = 0$. Cauchy–Schwarz: $1 = \text{Cov}(\hat{\theta}, U)^2 \leq \text{Var}(\hat{\theta}) \text{Var}(U) = \text{Var}(\hat{\theta}) nI(\theta)$. $\square$

Remark. For a biased estimator with $b(\theta) = \text{bias}(\hat{\theta})$ the same argument gives $\text{Var}(\hat{\theta}) \geq (1 + b'(\theta))^2 / (nI(\theta))$. Equality in Cauchy–Schwarz holds iff $\hat{\theta} - \theta$ is proportional to the score, which happens exactly in exponential families for the mean-value parameter.

5.2 Sufficiency (brief)

A statistic $T(X)$ is sufficient for θ if the conditional law of the sample given T does not depend on θ . By the Fisher–Neyman factorization theorem, T is sufficient iff $f_{n(x; \theta)} = g(T(x); \theta)h(x)$. For normal data with known σ , $\sum_i X_i$ is sufficient for μ ; for the Bernoulli model, $\sum_i X_i$ is sufficient for p . The Rao–Blackwell theorem states that conditioning an estimator on a sufficient statistic never increases its MSE.

6 Maximum likelihood

In the notes: Point Estimation and Maximum Likelihood

DEFINITION The likelihood is $L(\theta) = \prod_{i=1}^n f(x_i; \theta)$, the log-likelihood $\ell(\theta) = \log L(\theta)$ and the maximum likelihood estimator $\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} \ell(\theta)$.

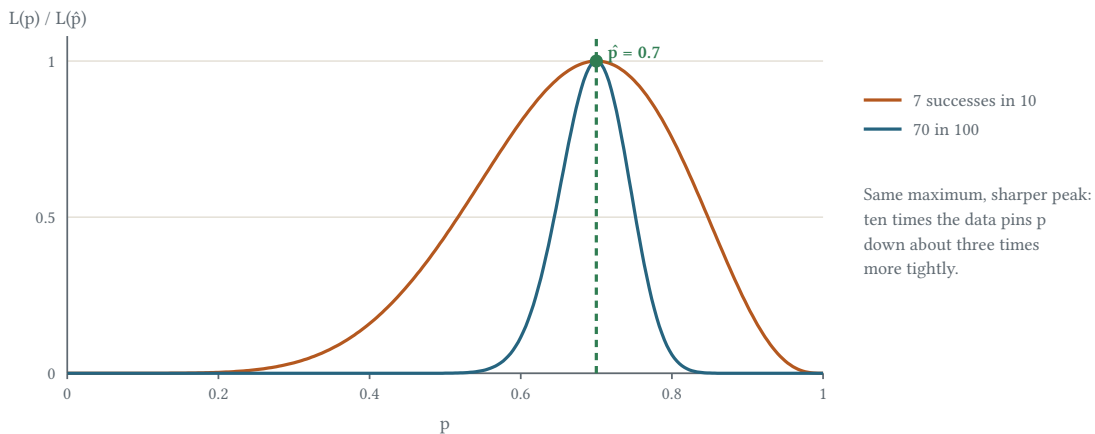


Figure 6: Likelihood of a success probability for the same observed proportion at two sample sizes, each scaled to peak at 1. The maximizer is the same; the curvature at the peak, which is the Fisher information, grows with n .

EXAMPLE Bernoulli $\ell(p) = (\sum x_i) \log p + (n - \sum x_i) \log(1 - p)$, $\ell'(p) = (\sum x_i)/p - (n - \sum x_i)/(1 - p)$. Setting it to zero gives $\hat{p} = \bar{x}$; $\ell'' < 0$ so it is a maximum. The information is $I(p) = 1/(p(1 - p))$ and $\text{Var}(\hat{p}) = p(1 - p)/n = 1/(nI(p))$: the MLE attains the Cramér–Rao bound exactly.

EXAMPLE Normal $\ell(\mu, \sigma^2) = -(n/2) \log(2\pi\sigma^2) - (2\sigma^2)^{-1} \sum_i (x_i - \mu)^2$. The first-order conditions give $\hat{\mu} = \bar{x}$ and $\hat{\sigma}^2 = n^{-1} \sum_i (x_i - \bar{x})^2$. By the theorem on S^2 , $\mathbb{E} \hat{\sigma}^2 = ((n - 1)/n)\sigma^2$: the MLE of the variance is biased, but consistent, and $\hat{\sigma}^2 = cS^2$ with $c = (n - 1)/n \rightarrow 1$.

6.1 Asymptotic properties

THEOREM Asymptotics of the MLE Under the regularity conditions above, with θ_0 the true parameter and $I(\theta_0) > 0$: (i) $\hat{\theta}_{\text{MLE}} \xrightarrow{P} \theta_0$; (ii) $\sqrt{n}(\hat{\theta}_{\text{MLE}} - \theta_0) \xrightarrow{d} \mathcal{N}(0, I(\theta_0)^{-1})$; (iii) invariance: the MLE of $g(\theta)$ is $g(\hat{\theta}_{\text{MLE}})$, and by the delta method it is asymptotically $\mathcal{N}(g(\theta_0), g'(\theta_0)^2/(nI(\theta_0)))$.

Proof. Sketch of (ii). Expand the score at the MLE around θ_0 : $0 = \ell'(\hat{\theta}) = \ell'(\theta_0) + (\hat{\theta} - \theta_0)\ell''(\tilde{\theta})$ for some $\tilde{\theta}$ between them. Hence

$$\sqrt{n}(\hat{\theta} - \theta_0) = \frac{n^{-1/2}\ell'(\theta_0)}{-n^{-1}\ell''(\tilde{\theta})}. \quad (14)$$

The numerator is $n^{-1/2} \sum_i s(X_i; \theta_0)$, a normalized sum of i.i.d. mean-zero variables with variance $I(\theta_0)$, so by the CLT it tends to $\mathcal{N}(0, I(\theta_0))$. By the law of large numbers and consistency, the denominator tends to $-E[\partial_{\theta}^2 \log f] = I(\theta_0)$. Slutsky gives the limit $\mathcal{N}(0, I(\theta_0)^{-1})$. $\square$

The practical consequence is the standard error $\hat{\text{se}} = 1/\sqrt{nI(\hat{\theta})}$ (or its observed-information version, $1/\sqrt{-\ell''(\hat{\theta})}$), valid for large n .

6.2 Method of moments

Equate the first d population moments $m_{j(\theta)} = E_{\theta} X^j$ to the sample moments $n^{-1} \sum_i X_i^j$ and solve for $\theta \in \mathbb{R}^d$. The estimator is consistent when $m = (m_1, \dots, m_d)$ is invertible with continuous inverse, but generally has larger asymptotic variance than the MLE.

7 Interval estimation

In the notes: Confidence Intervals

DEFINITION A $(1 - \alpha)$ confidence interval is a random interval $[L(X), U(X)]$ with $P_{\theta}(L(X) \leq \theta \leq U(X)) \geq 1 - \alpha$ for all $\theta \in \Theta$. The left side is the coverage; the inequality is replaced by an equality for exact intervals.

The probability refers to the random endpoints under repeated sampling. For the realized interval, θ is fixed and the event either holds or not.

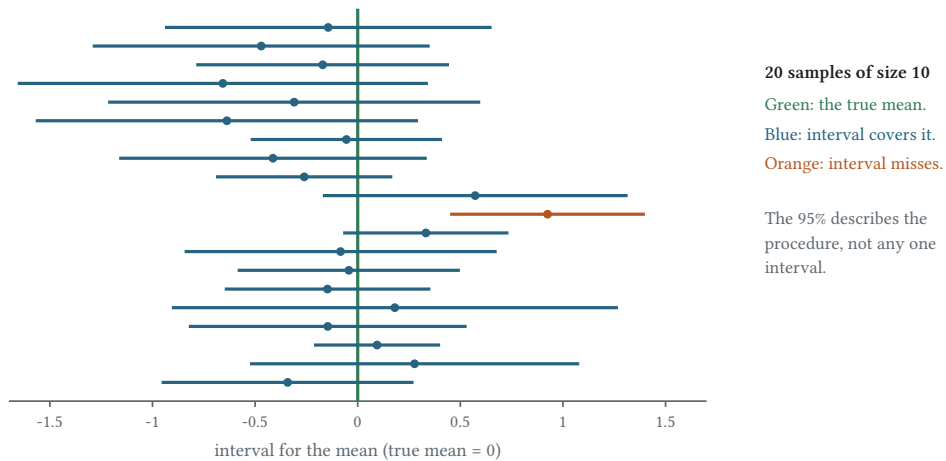


Figure 7: Twenty independent samples, each with its own 95% interval for the same true mean. The intervals vary, the truth does not; about one in twenty misses, here the orange one.

WATCHOUT The 95% describes the procedure, not the one interval you computed: it is not the probability that θ lies inside it.

DEFINITION Pivot $Q(X; \theta)$ is a pivot if its distribution does not depend on θ .

If $P(a \leq Q(X; \theta) \leq b) = 1 - \alpha$ and Q is monotone in θ , solving $a \leq Q \leq b$ for θ gives an exact interval.

THEOREM Intervals for the normal model Let $X_i \sim \mathcal{N}(\mu, \sigma^2)$ i.i.d.. Exact $(1 - \alpha)$ intervals:

- mean, σ unknown: $\bar{X} \pm t_{n-1, 1-\alpha/2} S / \sqrt{n}$;
- mean, σ known: $\bar{X} \pm z_{1-\alpha/2} \sigma / \sqrt{n}$;
 $[(n-1)S^2 / \chi_{n-1, 1-\alpha/2}^2, (n-1)S^2 / \chi_{n-1, \alpha/2}^2]$
- variance: $(n-1)S^2 / \chi_{n-1, \alpha/2}^2$.

Proof. $P(-t \leq T \leq t) = 1 - \alpha$ with $t = t_{n-1, 1-\alpha/2}$ and T the pivot of the sampling-distribution theorem; rearranging $-t \leq (\bar{X} - \mu) / (S / \sqrt{n}) \leq t$ gives the first interval. The second is identical with T replaced by $\sqrt{n}(\bar{X} - \mu) / \sigma \sim \mathcal{N}(0, 1)$. For the third, $P(\chi_{n-1, \alpha/2}^2 \leq (n-1)S^2 / \sigma^2 \leq \chi_{n-1, 1-\alpha/2}^2) = 1 - \alpha$ and inverting in σ^2 reverses the inequalities. $\square$

THEOREM Wald interval If $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} \mathcal{N}(0, v(\theta))$ with v continuous, then $\hat{\theta} \pm z_{1-\alpha/2} \sqrt{v(\hat{\theta})/n}$ has asymptotic coverage $1 - \alpha$. For the MLE, $v = I^{-1}$. For a proportion, $\hat{p} \pm z_{1-\alpha/2} \sqrt{\hat{p}(1-\hat{p})/n}$.

Proof. $\sqrt{n}(\hat{\theta} - \theta) / \sqrt{v(\hat{\theta})} = [\sqrt{n}(\hat{\theta} - \theta) / \sqrt{v(\theta)}] \cdot [\sqrt{v(\theta)/v(\hat{\theta})}]$; the first factor tends in distribution to $\mathcal{N}(0, 1)$ and the second in probability to 1 by consistency and continuity. Apply Slutsky. $\square$

Remark. The Wald interval for a proportion has coverage that oscillates and can be far below $1 - \alpha$ when n is small or p is near 0 or 1. The Wilson interval, obtained by inverting the score test instead of replacing p by $\hat{p}$ in the variance, is the standard fix.

8 Hypothesis testing

In the notes: Hypothesis Tests

8.1 Framework

A test of $H_0 : \theta \in \Theta_0$ against $H_1 : \theta \in \Theta_1$ is a rule $\varphi(X) \in \{0, 1\}$ (1 means reject), typically $\varphi = \mathbb{1}\{T(X) \in R\}$ for a statistic T and rejection region R .

DEFINITION The size of the test is $\sup_{\theta \in \Theta_0} P_\theta(\varphi = 1)$; it has level α if the size is at most α . The power function is $\beta(\theta) = P_\theta(\varphi = 1)$, so the Type II error at $\theta \in \Theta_1$ is $1 - \beta(\theta)$.

DEFINITION The p -value of observed t_{obs} for a statistic T that rejects for large values is $p = \sup_{\theta \in \Theta_0} P_\theta(T \geq t_{\text{obs}})$.

LEMMA Uniformity of the p -value If H_0 is simple, T has a continuous distribution function G under H_0 , and rejection is for large T , then $p(T) = 1 - G(T) \sim \text{Uniform}(0, 1)$ under H_0 . Hence rejecting when $p \leq \alpha$ has size exactly α .

Proof. $G(T) \sim \text{Uniform}(0, 1)$ by the probability integral transform: $P(G(T) \leq u) = P(T \leq G^{-1}(u)) = u$. Then $1 - G(T)$ is also uniform. $\square$

8.2 The t -tests

THEOREM One-sample and Welch t -tests (i) For $X_i \sim \mathcal{N}(\mu, \sigma^2)$ i.i.d., under $H_0 : \mu = \mu_0$ the statistic $T = (\bar{X} - \mu_0)/(S/\sqrt{n})$ has the t_{n-1} distribution, so rejecting for $|T| > t_{n-1, 1-\alpha/2}$ has size α . (ii) For two independent normal samples with means μ_1, μ_2 , and $H_0 : \mu_1 = \mu_2$,

$$T = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{S_1^2/n_1 + S_2^2/n_2}} \quad (15)$$

is approximately t_ν with

$$\nu = \frac{(S_1^2/n_1 + S_2^2/n_2)^2}{(S_1^2/n_1)^2/(n_1 - 1) + (S_2^2/n_2)^2/(n_2 - 1)}. \quad (16)$$

Part (i) is the pivot theorem with $\mu = \mu_0$. In (ii) the denominator is a weighted sum of independent scaled χ^2 variables, which is not exactly a scaled χ^2 ; the Welch-Satterthwaite ν matches the first two moments of a scaled χ_ν^2 to it. With equal variances the pooled test is exactly $t_{n_1+n_2-2}$, but Welch is robust to unequal variances at little cost.

THEOREM Duality of tests and intervals If, for each θ_0 , $A(\theta_0)$ is the acceptance region of a level- α test of $\theta = \theta_0$, then $C(x) = \{\theta_0 : x \in A(\theta_0)\}$ is a $(1 - \alpha)$ confidence set. Conversely, from a confidence set one obtains a test by rejecting $\theta = \theta_0$ iff $\theta_0 \notin C(x)$.

Proof. $P_{\theta_0}(\theta_0 \in C(X)) = P_{\theta_0}(X \in A(\theta_0)) \geq 1 - \alpha$, for every θ_0 . The converse is the same identity read backwards. $\square$

8.3 Power and sample size

THEOREM Power of the one-sided z -test Let $X_i \sim \mathcal{N}(\mu, \sigma^2)$ with σ known, and reject $H_0 : \mu = \mu_0$ in favour of $H_1 : \mu > \mu_0$ when $Z = \sqrt{n}(\bar{X} - \mu_0)/\sigma > z_{1-\alpha}$. Then

$$\beta(\mu_1) = 1 - \Phi(z_{1-\alpha} - \sqrt{n}(\mu_1 - \mu_0)/\sigma), \quad (17)$$

and $\beta(\mu_1) \geq 1 - \beta_0$ holds iff $n \geq \left((z_{1-\alpha} + z_{1-\beta_0})\sigma/(\mu_1 - \mu_0)\right)^2$.

Proof. Under $\mu = \mu_1$, $\sqrt{n}(\bar{X} - \mu_1)/\sigma \sim \mathcal{N}(0, 1)$, so $Z = Z_0 + \sqrt{n}(\mu_1 - \mu_0)/\sigma$ with $Z_0 \sim \mathcal{N}(0, 1)$. Hence $P(Z > z_{1-\alpha}) = 1 - \Phi(z_{1-\alpha} - \sqrt{n}(\mu_1 - \mu_0)/\sigma)$. Requiring this to be at least $1 - \beta_0$ means $z_{1-\alpha} - \sqrt{n}\delta/\sigma \leq -z_{1-\beta_0}$ (with $\delta = \mu_1 - \mu_0$), and solving for n gives the bound. $\square$

The required n scales like σ^2/δ^2 : halving the detectable effect quadruples the sample.

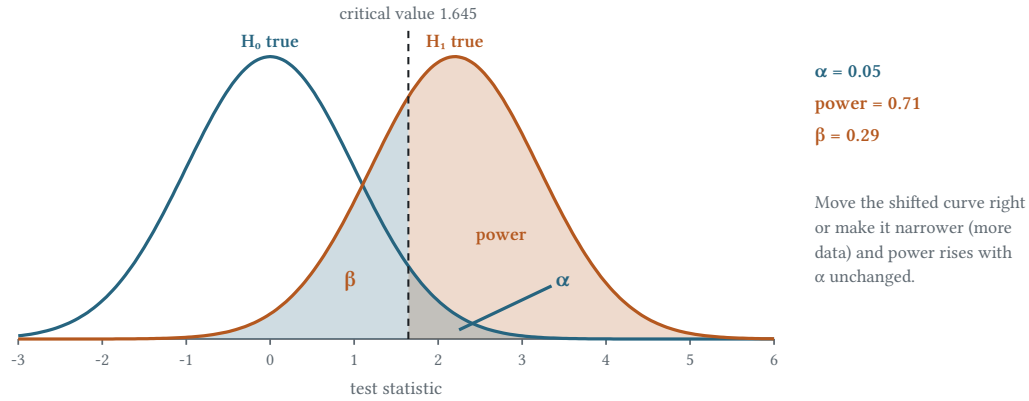


Figure 8: One-sided z -test at $\alpha = 0.05$ when the true mean is 2.2 standard errors above the null. The blue sliver beyond the line is the Type I error rate, the orange area to its right is the power, and the orange area to its left is the Type II error rate.

8.4 Likelihood ratio tests

THEOREM Neyman–Pearson lemma For simple $H_0 : \theta = \theta_0$ against simple $H_1 : \theta = \theta_1$, let φ^* reject when $L(\theta_1)/L(\theta_0) > k$, with k chosen so that φ^* has size α . Then for any other test φ of size at most α , $\beta_{\varphi}(\theta_1) \leq \beta_{\varphi^*}(\theta_1)$.

Proof. For every x , $(\varphi^*(x) - \varphi(x))(f(x; \theta_1) - kf(x; \theta_0)) \geq 0$: where $\varphi^* = 1$ the second factor is positive, where $\varphi^* = 0$ it is non-positive and the first factor is $-\varphi(x) \leq 0$ or zero. Integrating over x gives $\beta_{\varphi^*}(\theta_1) - \beta_{\varphi}(\theta_1) \geq k(\alpha_{\varphi^*} - \alpha_{\varphi}) \geq 0$, since both sizes are at most α and φ^* has size exactly α . $\square$

THEOREM Wilks' theorem Let $\Lambda = \sup_{\Theta_0} L / \sup_{\Theta} L$, where Θ has d free parameters and Θ_0 has $d - r$. Under the regularity conditions and H_0 ,

$$-2 \log \Lambda \xrightarrow{d} \chi_r^2. \quad (18)$$

The proof expands ℓ quadratically around the MLE and uses asymptotic normality of the MLE, so that $-2 \log \Lambda$ becomes a quadratic form in r asymptotically independent standard normals. The Wald and score tests are asymptotically equivalent to it. In the linear model with Gaussian errors, the likelihood ratio is a monotone function of an F statistic, which is why the F -test of regression has an exact distribution.

8.5 Multiple testing

With m tests and all nulls true, each at level α , the expected number of false rejections is $m\alpha$.

THEOREM Bonferroni and Benjamini–Hochberg (i) Rejecting each hypothesis with $p_i \leq \alpha/m$ controls the family-wise error rate, $P(\text{at least one false rejection}) \leq \alpha$, with no assumption on the dependence of the p -values. (ii) Order the p -values $p_{(1)} \leq \dots \leq p_{(m)}$ and reject the k smallest, where $k = \max\{i : p_{(i)} \leq i\alpha/m\}$. If the p -values are independent, the false discovery rate $E[V / \max(R, 1)]$ (with V false rejections out of R) is at most $\alpha m_0/m \leq \alpha$, where m_0 is the number of true nulls.

Proof. (i) By the union bound, $P(\bigcup_{i \in H_0} \{p_i \leq \alpha/m\}) \leq m_0 \alpha/m \leq \alpha$. (ii) is the Benjamini–Hochberg (1995) theorem; the proof conditions on the other p -values and is omitted. $\square$

9 Linear regression

In the notes: Linear Regression · Regression Diagnostics and Leverage

This chapter uses expectation, variance and covariance of random vectors, the multivariate linear map rule $\text{Var}(AY) = A \text{Var}(Y) A^\top$, and the χ^2 , t and F laws, all from the chapters above.

9.1 The model and least squares

Observe $\mathbf{y} \in \mathbb{R}^n$ and a fixed design matrix $X \in \mathbb{R}^{n \times p}$ of full column rank, usually with a column of ones. The linear model is

$$\mathbf{y} = X\beta + \epsilon, \quad \mathbb{E} \epsilon = 0, \quad \text{Var}(\epsilon) = \sigma^2 I_n. \quad (19)$$

DEFINITION Least squares $\hat{\beta} = \arg \min_{\beta} \|\mathbf{y} - X\beta\|^2$, and the residual sum of squares is $\text{RSS} = \|\mathbf{y} - X\hat{\beta}\|^2$.

THEOREM Normal equations $\hat{\beta}$ is the unique solution of $X^\top X \hat{\beta} = X^\top \mathbf{y}$, that is $\hat{\beta} = (X^\top X)^{-1} X^\top \mathbf{y}$.

Proof. $X^\top X$ is invertible because X has full column rank. Put $\mathbf{e} = \mathbf{y} - X\hat{\beta}$ with $\hat{\beta}$ the stated solution, so $X^\top \mathbf{e} = 0$. For any β , $\mathbf{y} - X\beta = \mathbf{e} + X(\hat{\beta} - \beta)$ and the two parts are orthogonal, hence $\|\mathbf{y} - X\beta\|^2 = \|\mathbf{e}\|^2 + \|X(\hat{\beta} - \beta)\|^2$, which is smallest, and only, at $\beta = \hat{\beta}$. $\square$

DEFINITION Hat matrix $H = X(X^\top X)^{-1} X^\top$, so that $\hat{\mathbf{y}} = H\mathbf{y}$ and $\mathbf{e} = (I - H)\mathbf{y}$.

LEMMA Projection H is symmetric and idempotent, $HX = X$, and $\text{tr } H = p$. Hence H is the orthogonal projection onto $\text{col}(X)$ and $I - H$ the projection onto its complement, of rank $n - p$. In particular $X^\top \mathbf{e} = 0$, and if X has an intercept column, $\sum_i e_i = 0$.

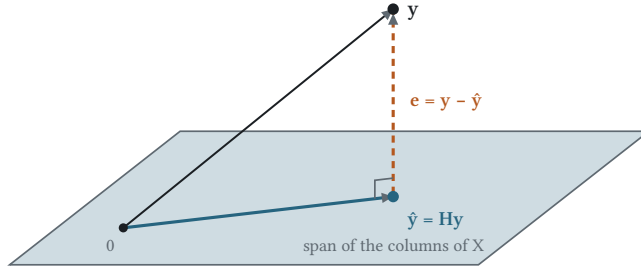
Proof. Symmetry and $H^2 = X(X^\top X)^{-1} X^\top X(X^\top X)^{-1} X^\top = H$ are direct, and $HX = X$ follows by cancelling. The trace of an idempotent matrix is its rank, and $\text{tr } H = \text{tr}((X^\top X)^{-1} X^\top X) = \text{tr } I_p = p$. Then $X^\top (I - H) = 0$; the intercept column gives $\mathbf{1}^\top \mathbf{e} = 0$. $\square$

LEMMA Moments of the estimates $\mathbb{E} \hat{\beta} = \beta$, $\text{Var}(\hat{\beta}) = \sigma^2 (X^\top X)^{-1}$, $\mathbb{E} \mathbf{e} = 0$, $\text{Var}(\mathbf{e}) = \sigma^2 (I - H)$, and $s^2 = \text{RSS} / (n - p)$ is unbiased for σ^2 .

Proof. $\hat{\beta} = A\mathbf{y}$ with $A = (X^\top X)^{-1} X^\top$ and $AX = I$, so $\mathbb{E} \hat{\beta} = \beta$, and $AA^\top = (X^\top X)^{-1}$ gives the variance. Also $\mathbf{e} = (I - H)\mathbf{e}$, so $\text{Var}(\mathbf{e}) = \sigma^2 (I - H)^2 = \sigma^2 (I - H)$. Then $\mathbb{E} \|\mathbf{e}\|^2 = \text{tr } \text{Var}(\mathbf{e}) = \sigma^2 (n - p)$. $\square$

THEOREM Gauss–Markov Among all linear unbiased estimators $\mathbf{a}^\top \mathbf{y}$ of $\mathbf{c}^\top \beta$, the least squares estimator $\mathbf{c}^\top \hat{\beta}$ has the smallest variance.

Proof. Unbiasedness for every β means $\mathbf{a}^\top X = \mathbf{c}^\top$. Write $\mathbf{a} = X(X^\top X)^{-1} \mathbf{c} + \mathbf{d}$; then $\mathbf{d}^\top X = 0$. The two parts of $\mathbf{a}$ are orthogonal, so $\text{Var}(\mathbf{a}^\top \mathbf{y}) = \sigma^2 \left(\|X(X^\top X)^{-1} \mathbf{c}\|^2 + \|\mathbf{d}\|^2 \right) \geq \sigma^2 \mathbf{c}^\top (X^\top X)^{-1} \mathbf{c} = \text{Var}(\mathbf{c}^\top \hat{\beta})$. $\square$



The fit is the point of the plane closest to y , so the residual is perpendicular to every column of X : $X^T e = 0$. These are the normal equations, and the coordinates of $\hat{y}$ in the columns are the coefficients.

Figure 9: Least squares as a projection. The fitted vector $\hat{y}$ is the point of $\text{col}(X)$ nearest to y , the residual is orthogonal to every column, and the coefficients are the coordinates of $\hat{y}$ in the columns.

9.2 Normal errors and inference

THEOREM Sampling distributions If in addition $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$, then $\hat{\beta} \sim \mathcal{N}(\beta, \sigma^2 (X^T X)^{-1})$, $(n - p)s^2 / \sigma^2 \sim \chi_{n-p}^2$, and $\hat{\beta}$ is independent of s^2 . Under these assumptions $\hat{\beta}$ is also the maximum likelihood estimator, and the MLE of σ^2 is RSS / n .

Proof. $\hat{\beta}$ and e are linear in y , hence jointly normal, with $\text{Cov}(\hat{\beta}, e) = \sigma^2 A(I - H) = 0$ since $AH = A$; uncorrelated jointly normal vectors are independent. In an orthonormal basis whose last $n - p$ vectors span $\text{col}(X)^\perp$, $\|e\|^2 / \sigma^2$ is the sum of the squares of the last $n - p$ coordinates of ϵ / σ , which are i.i.d. $\mathcal{N}(0, 1)$. The likelihood is maximized in β by minimizing $\|y - X\beta\|^2$. $\square$

THEOREM t statistics and intervals Under normal errors, with $\hat{\text{se}}(\hat{\beta}_j) = s \sqrt{[(X^T X)^{-1}]_{jj}}$,

$$(\hat{\beta}_j - \beta_j) / \hat{\text{se}}(\hat{\beta}_j) \sim t_{n-p}. \quad (20)$$

The interval $\hat{\beta}_j \pm t_{n-p, 1-\alpha/2} \hat{\text{se}}(\hat{\beta}_j)$ has coverage $1 - \alpha$, and the t -test of $\beta_j = 0$ rejects when $|\hat{\beta}_j| / \hat{\text{se}}$ exceeds the same quantile. At a new point x_0 the mean response $x_0^T \beta$ has variance $\sigma^2 x_0^T (X^T X)^{-1} x_0$ for its estimate, while a new observation is predicted with error variance $\sigma^2 (1 + x_0^T (X^T X)^{-1} x_0)$.

Proof. Standardize the normal $\hat{\beta}_j$ and divide by the independent $\sqrt{\chi_{n-p}^2 / (n - p)}$: this is the definition of t_{n-p} . The prediction variance adds the variance σ^2 of the new error to that of $x_0^T \hat{\beta}$. $\square$

9.3 Sums of squares and the F -test

DEFINITION Sums of squares and R^2 $\text{SST} = \sum_i (y_i - \bar{y})^2$, $\text{SSR} = \sum_i (\hat{y}_i - \bar{y})^2$, $\text{SSE} = \text{RSS}$, and $R^2 = \text{SSR} / \text{SST}$. The adjusted version is $1 - (\text{SSE} / (n - p)) / (\text{SST} / (n - 1))$.

THEOREM Decomposition With an intercept, $\text{SST} = \text{SSR} + \text{SSE}$, and $R^2 = \text{corr}(y, \hat{y})^2 = 1 - \text{SSE} / \text{SST}$.

Proof. $y - \bar{y}\mathbf{1} = (\hat{y} - \bar{y}\mathbf{1}) + e$, the first part lies in $\text{col}(X)$ (it contains $\mathbf{1}$) and e is orthogonal to it, so Pythagoras applies. Moreover $\sum_i (y_i - \bar{y})(\hat{y}_i - \bar{y}) = \text{SSR} + \sum_i e_i(\hat{y}_i - \bar{y}) = \text{SSR}$, so the squared correlation is $\text{SSR}^2 / (\text{SST} \cdot \text{SSR})$, the same ratio. $\square$

THEOREM F -test of nested models Let the reduced model with $p - q$ parameters have column space inside $\text{col}(X)$ and residual sum of squares $\text{SSE}_0 \geq \text{SSE}$. Under normal errors and the reduced model,

$$F = \frac{(\text{SSE}_0 - \text{SSE})/q}{\text{SSE}/(n-p)} \sim F_{q, n-p}. \quad (21)$$

The overall test of “no predictor matters” has $q = p - 1$ and $F = (\text{SSR}/(p-1))/(\text{SSE}/(n-p))$. For $q = 1$, $F = t^2$ for the corresponding coefficient.

Proof. $\text{SSE}_0 - \text{SSE} = \|(H - H_0)\mathbf{y}\|^2$ where H_0 projects onto the smaller space, and $H - H_0$ is a projection of rank q orthogonal to $I - H$. Under the reduced model $(H - H_0)\mathbf{y} = (H - H_0)\boldsymbol{\epsilon}$, so the two sums of squares divided by σ^2 are independent χ_q^2 and χ_{n-p}^2 . $\square$

SOURCE	SUM OF SQUARES	DF AND MEAN SQUARE
Regression	SSR	$p - 1, \text{SSR}/(p - 1)$
Residual	SSE	$n - p, s^2 = \text{SSE}/(n - p)$
Total	SST	$n - 1$

9.4 Leverage, influence and collinearity

LEMMA Leverage Let $h_{ii} = \mathbf{x}_i^\top (X^\top X)^{-1} \mathbf{x}_i$ be the diagonal of H . Then $\text{Var}(e_i) = \sigma^2(1 - h_{ii})$, $\sum_i h_{ii} = p$, $0 \leq h_{ii} \leq 1$, and $h_{ii} \geq 1/n$ with an intercept. In simple regression $h_{ii} = 1/n + (x_i - \bar{x})^2/S_{xx}$. The fitted value is $\hat{y}_i = h_{ii}y_i + \sum_{j \neq i} h_{ij}y_j$.

Proof. The variance is the diagonal of $\sigma^2(I - H)$ and the trace is $\text{tr } H = p$. From $H^2 = H$, $h_{ii} = \sum_j h_{ij}^2 \geq h_{ii}^2$, so $h_{ii} \in [0, 1]$. If $\mathbf{1} \in \text{col}(X)$ then $H - \mathbf{1}\mathbf{1}^\top/n$ is again a projection, so its diagonal is nonnegative. $\square$

DEFINITION Studentized residuals The internally studentized residual is $r_i = e_i/(s\sqrt{1 - h_{ii}})$. The externally studentized residual replaces s by $s_{(i)}$, computed without observation i ; under normal errors it has a t_{n-p-1} distribution.

LEMMA Deleting one observation With $M = X^\top X$ and $\hat{\beta}_{(i)}$ the estimate without observation i ,

$$\hat{\beta} - \hat{\beta}_{(i)} = \frac{M^{-1}\mathbf{x}_i e_i}{1 - h_{ii}}, \quad y_i - \mathbf{x}_i^\top \hat{\beta}_{(i)} = \frac{e_i}{1 - h_{ii}}. \quad (22)$$

Proof. By Sherman–Morrison, $(M - \mathbf{x}_i \mathbf{x}_i^\top)^{-1} = M^{-1} + M^{-1}\mathbf{x}_i \mathbf{x}_i^\top M^{-1}/(1 - h_{ii})$, and the deleted cross-product is $X^\top \mathbf{y} - \mathbf{x}_i y_i$. Multiplying out, $\hat{\beta}_{(i)} = \hat{\beta} - M^{-1}\mathbf{x}_i y_i + M^{-1}\mathbf{x}_i(\hat{y}_i - h_{ii}y_i)/(1 - h_{ii}) = \hat{\beta} - M^{-1}\mathbf{x}_i e_i/(1 - h_{ii})$. For the second identity, $y_i - \mathbf{x}_i^\top \hat{\beta}_{(i)} = e_i + h_{ii}e_i/(1 - h_{ii})$. $\square$

DEFINITION Cook's distance $D_i = (\hat{\beta} - \hat{\beta}_{(i)})^\top X^\top X(\hat{\beta} - \hat{\beta}_{(i)})/(ps^2) = \|\hat{\mathbf{y}} - \hat{\mathbf{y}}_{(i)}\|^2/(ps^2)$.

THEOREM Cook's distance, closed form $D_i = \frac{r_i^2}{p} \cdot \frac{h_{ii}}{1 - h_{ii}}$.

Proof. Substitute the deletion formula: $D_i = e_i^2 \mathbf{x}_i^\top M^{-1} M M^{-1} \mathbf{x}_i / ((1 - h_{ii})^2 ps^2) = e_i^2 h_{ii} / (ps^2(1 - h_{ii})^2)$, which is the stated product. $\square$

Remark. The second identity of the deletion lemma gives the leave-one-out residuals without refitting, $\text{PRESS} = \sum_i (e_i/(1 - h_{ii}))^2$. Common flags are $h_{ii} > 2p/n$ and $D_i > 4/n$ (or > 1); they are screening devices, not tests.

THEOREM Variance inflation Let R_j^2 be the R^2 of the regression of column j on the other columns, and $S_{jj} = \sum_i (x_{ij} - \bar{x}_j)^2$. Then

$$\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{(1 - R_j^2)S_{jj}} = \text{VIF}_j \cdot \frac{\sigma^2}{S_{jj}}, \quad \text{VIF}_j = \frac{1}{1 - R_j^2}. \quad (23)$$

Proof. Let M_{-j} be the projection onto $\text{col}(X_{-j})^\perp$ and $\tilde{x}_j = M_{-j}x_j$. Applying M_{-j} to $y = \hat{y} + e$ kills the other columns and fixes e , so $M_{-j}y = \tilde{x}_j\hat{\beta}_j + e$ and, since $e \perp \tilde{x}_j$, $\hat{\beta}_j = \tilde{x}_j^\top y / \|\tilde{x}_j\|^2$ (Frisch-Waugh-Lovell). Then $\text{Var}(\hat{\beta}_j) = \sigma^2 / \|\tilde{x}_j\|^2$, and $\|\tilde{x}_j\|^2 = (1 - R_j^2)S_{jj}$ is the residual sum of squares of that auxiliary regression. $\square$

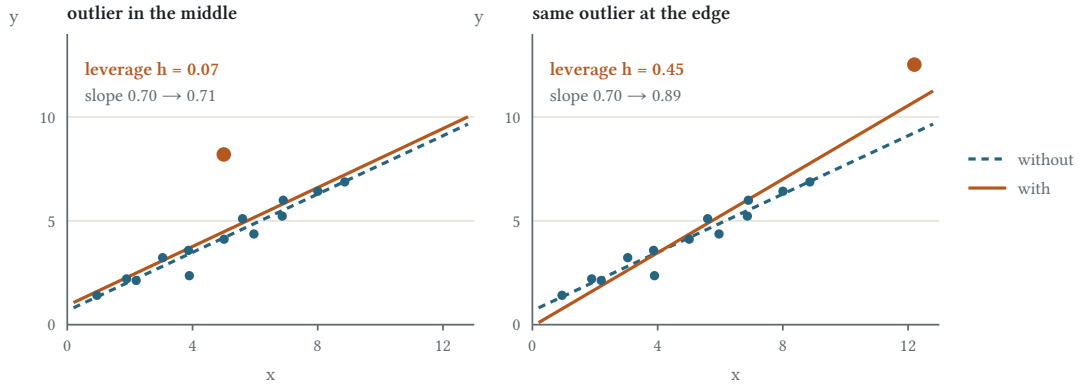


Figure 10: The same vertical outlier in the middle and at the edge of the x range. Dashed: fit without it; solid: fit with it. Leverage, not the size of the residual alone, decides how far the line moves.

10 Regularization

In the notes: Regularization: Ridge and Lasso

Notation as in the previous chapter; the columns of X are centred and scaled, the intercept is not penalized, and $\lambda \geq 0$ is the penalty weight.

10.1 Prediction error and the trade-off

THEOREM Bias-variance decomposition of prediction error Let $y_0 = f(x_0) + \epsilon_0$ be a new observation, independent of the training data, with $E\epsilon_0 = 0$ and $\text{Var}(\epsilon_0) = \sigma^2$, and let $\hat{f}$ be fitted on the training data. Then

$$E[(y_0 - \hat{f}(x_0))^2] = \sigma^2 + (E\hat{f}(x_0) - f(x_0))^2 + \text{Var}(\hat{f}(x_0)). \quad (24)$$

Proof. Write $y_0 - \hat{f} = \epsilon_0 + (f - E\hat{f}) + (E\hat{f} - \hat{f})$. The three terms are independent of each other or have zero mean, so the cross terms vanish, and squaring gives irreducible noise, squared bias and variance. $\square$

10.2 Ridge regression

DEFINITION Ridge regression $\hat{\beta}_\lambda = \arg \min_\beta \|y - X\beta\|^2 + \lambda\|\beta\|^2 = (X^\top X + \lambda I)^{-1} X^\top y$.

With the singular value decomposition $X = UDV^\top$ and singular values $d_1 \geq \dots \geq d_p > 0$, in the coordinates $\gamma = V^\top \beta$ the estimate shrinks each least squares coordinate by a factor,

$$\hat{\gamma}_{\lambda,j} = \frac{d_j^2}{d_j^2 + \lambda} \hat{\gamma}_j. \quad (25)$$

The fitted values are $\hat{\mathbf{y}}_\lambda = H_\lambda \mathbf{y}$ with $H_\lambda = X(X^\top X + \lambda I)^{-1} X^\top$.

DEFINITION Effective degrees of freedom For a linear fit $\hat{\mathbf{y}} = S\mathbf{y}$ the effective degrees of freedom are $\text{df} = \text{tr } S$. For ridge, $\text{df}(\lambda) = \sum_j d_j^2 / (d_j^2 + \lambda)$, decreasing from p at $\lambda = 0$ to 0.

THEOREM Ridge can beat least squares The coordinate j of the ridge estimate has mean squared error

$$\text{MSE}_j(\lambda) = \frac{\sigma^2 d_j^2 + \lambda^2 \gamma_j^2}{(d_j^2 + \lambda)^2}, \quad (26)$$

and $\text{MSE}_j'(0) = -2\sigma^2/d_j^4 < 0$. Hence for every true β there is $\lambda > 0$ at which the total $\mathbb{E} \|\hat{\beta}_\lambda - \beta\|^2$ is smaller than that of least squares.

Proof. By the moment lemma, $\hat{\gamma}_j$ has mean γ_j and variance σ^2/d_j^2 . Scaling by $d_j^2/(d_j^2 + \lambda)$ gives variance $\sigma^2 d_j^2 / (d_j^2 + \lambda)^2$ and bias $-\lambda \gamma_j / (d_j^2 + \lambda)$. Differentiate the sum at $\lambda = 0$, where the numerator's derivative vanishes. $\square$

Remark. Ridge is the posterior mean, and mode, under $\mathbf{y} \mid \beta \sim \mathcal{N}(X\beta, \sigma^2 I)$ and prior $\beta \sim \mathcal{N}(0, \tau^2 I)$ with $\lambda = \sigma^2/\tau^2$, because $-2\sigma^2 \log \text{posterior} = \|\mathbf{y} - X\beta\|^2 + (\sigma^2/\tau^2)\|\beta\|^2 + \text{const.}$

10.3 Lasso

DEFINITION Lasso $\hat{\beta}_\lambda = \arg \min_\beta \frac{1}{2} \|\mathbf{y} - X\beta\|^2 + \lambda \|\beta\|_1$.

The problem is convex, so a minimizer exists, but it need not be unique unless the columns are in general position. Subgradient optimality gives the conditions below.

THEOREM Optimality conditions and soft-thresholding $\hat{\beta}$ is a minimizer iff, for every j , $X_j^\top (\mathbf{y} - X\hat{\beta}) = \lambda \text{sign}(\hat{\beta}_j)$ if $\hat{\beta}_j \neq 0$, and $|X_j^\top (\mathbf{y} - X\hat{\beta})| \leq \lambda$ if $\hat{\beta}_j = 0$. In particular $\hat{\beta} = 0$ iff $\lambda \geq \|X^\top \mathbf{y}\|_\infty$. If $X^\top X = I$, with b_j the least squares coefficients,

$$\hat{\beta}_{\lambda,j} = \text{sign}(b_j) (|b_j| - \lambda)_+. \quad (27)$$

Proof. The objective is convex, so 0 must lie in the subdifferential $-X^\top (\mathbf{y} - X\beta) + \lambda \partial \|\beta\|_1$, which is the stated pair of conditions. For $X^\top X = I$ the objective separates into $\frac{1}{2} (b_j - \beta_j)^2 + \lambda |\beta_j|$ plus a constant; setting the subgradient to zero gives $\beta_j = b_j - \lambda \text{sign}(\beta_j)$ when this has the sign of b_j , and $\beta_j = 0$ when $|b_j| \leq \lambda$. $\square$

RULE	PENALTY	ESTIMATE FOR ORTHONORMAL COLUMNS
Least squares	none	b_j
Ridge	$\lambda \ \beta\ ^2$	$b_j / (1 + \lambda)$
Lasso	$\lambda \ \beta\ _1$	$\text{sign}(b_j) (b_j - \lambda)_+$
Best subset	$\lambda \ \beta\ _0$ (with $\frac{1}{2} \ \mathbf{y} - X\beta\ ^2$)	$b_j \mathbb{1}\{ b_j > \sqrt{2\lambda}\}$

Remark. The lasso solution has at most $\min(n, p)$ nonzero coefficients. It is the MAP estimate under independent Laplace priors. The elastic net adds the ridge term, $\frac{1}{2} \|\mathbf{y} - X\beta\|^2 + \lambda (\alpha \|\beta\|_1 + \frac{1-\alpha}{2} \|\beta\|^2)$, which

keeps the sparsity and spreads weight over correlated predictors. λ is chosen by cross-validation, over a grid starting from $\|X^\top y\|_\infty$.

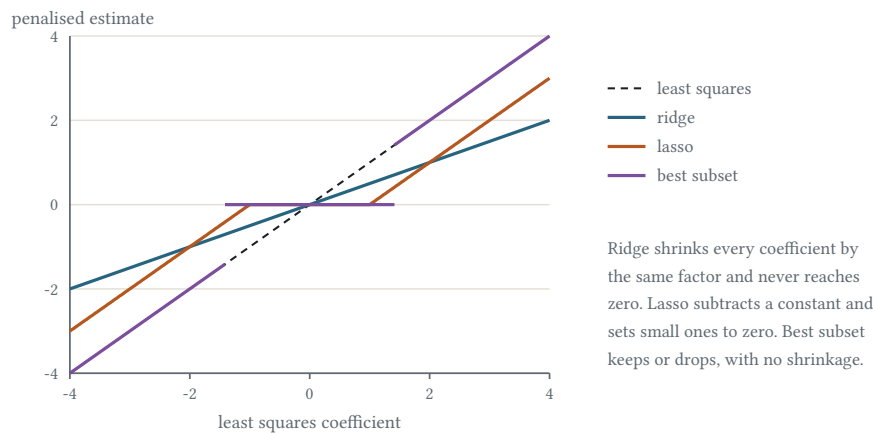


Figure 11: With orthonormal predictors the three penalties are three functions of the least squares coefficient: proportional shrinkage (ridge), soft thresholding (lasso) and hard thresholding (best subset).

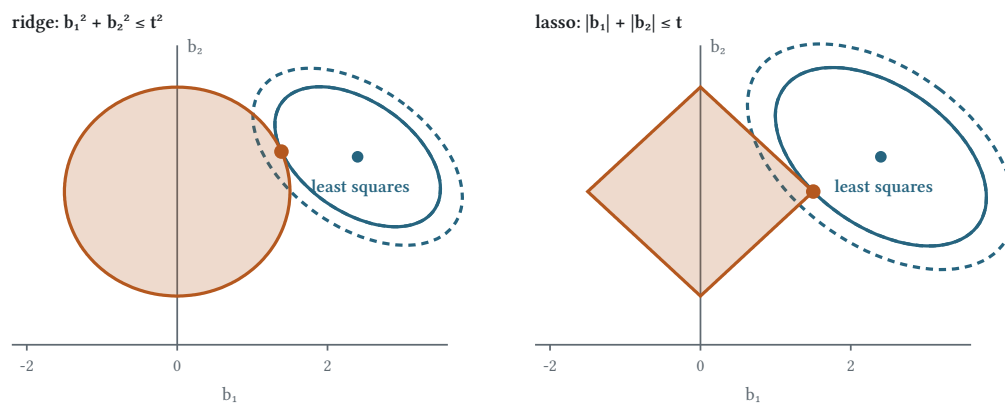


Figure 12: Elliptical contours of the residual sum of squares around the least squares estimate meet the constraint region. The disc has no corners, so ridge lands in the open; the diamond has corners on the axes, where a coefficient is exactly zero.

11 Splines and additive models

In the notes: Regression and Smoothing Splines · GAMs and MARS

11.1 Basis expansions and regression splines

DEFINITION Basis expansion $f(x) = \sum_{m=1}^M \beta_m h_m(x)$ for fixed functions h_m . The model is linear in β , so everything in the linear regression chapter applies with $X_{im} = h_m(x_i)$.

DEFINITION Cubic spline With knots $\xi_1 < \dots < \xi_K$, a cubic spline is a piecewise cubic polynomial that is twice continuously differentiable at every knot. A natural cubic spline is in addition linear on $(-\infty, \xi_1]$ and $[\xi_K, \infty)$.

THEOREM Dimension and bases The cubic splines with knots $\xi_1, \dots, \xi_K$ form a vector space of dimension $K + 4$ with the truncated power basis $1, x, x^2, x^3, (x - \xi_k)_+^3$ for $k = 1, \dots, K$. The natural cubic splines form a subspace of dimension K (for $K \geq 2$).

Proof. A twice differentiable gluing at ξ_k allows a jump only in the third derivative, which is exactly what adding $c_k(x - \xi_k)_+^3$ to a cubic does, and every spline has this form. Requiring zero second and third derivative on each of the two outer intervals gives four independent linear conditions on the $K + 4$ coefficients, leaving K . $\square$

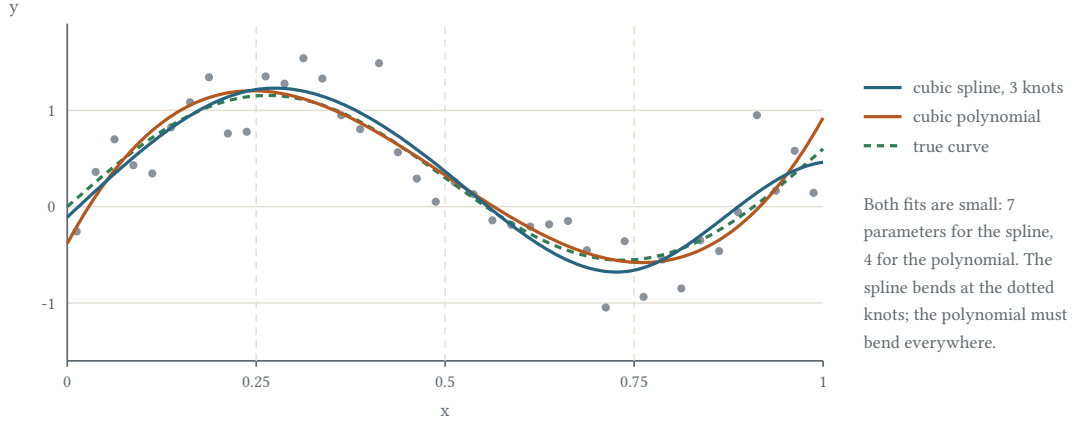


Figure 13: A cubic polynomial and a cubic spline with three knots fitted to the same noisy data. The spline has seven parameters and follows the curve; the global polynomial cannot.

11.2 Smoothing splines

DEFINITION Smoothing spline For $\lambda > 0$, $\hat{f} = \arg \min_f \sum_i (y_i - f(x_i))^2 + \lambda \int f''(t)^2 dt$ over twice differentiable f with f'' square integrable.

THEOREM Reinsch For distinct $x_1 < \dots < x_n$ the minimizer is the natural cubic spline with knots at the x_i , and its values at the knots are $\hat{\mathbf{f}} = (I + \lambda K)^{-1} \mathbf{y}$, where K is the positive semidefinite matrix with $\int f''^2 = \mathbf{f}^\top K \mathbf{f}$ for the natural spline through $\mathbf{f}$.

Proof. Let g be any admissible function and f the natural cubic spline with $f(x_i) = g(x_i)$. Put $h = g - f$, so $h(x_i) = 0$. Integrating by parts twice, $\int f'' h'' = [f'' h']_{x_1}^{x_n} - \int f''' h' = -\sum_i f'''(x_i^+) (h(x_{i+1}) - h(x_i)) = 0$, using $f''' = 0$ outside $[x_1, x_n]$, f''' piecewise constant and h vanishing at the knots. Hence $\int g''^2 = \int f''^2 + \int h''^2 \geq \int f''^2$, with equality only for $h'' = 0$, i.e. $g = f$. The fit terms agree, so the minimum is over natural splines, where the criterion is $\|\mathbf{y} - \mathbf{f}\|^2 + \lambda \mathbf{f}^\top K \mathbf{f}$, minimized at the stated $\hat{\mathbf{f}}$. $\square$

DEFINITION Linear smoother A fit of the form $\hat{\mathbf{y}} = S \mathbf{y}$. For the smoothing spline $S_\lambda = (I + \lambda K)^{-1}$ is symmetric with eigenvalues $1/(1 + \lambda d_k) \in (0, 1]$, d_k the eigenvalues of K ; the constants and the linear functions span its null space, so two eigenvalues equal 1 for every λ . Thus $\text{df}(\lambda) = \text{tr } S_\lambda = \sum_k 1/(1 + \lambda d_k)$ decreases from n to 2. Leave-one-out residuals obey $y_i - \hat{f}_{(i)}(x_i) = (y_i - \hat{f}(x_i))/(1 - S_{ii})$, and generalized cross-validation is $\text{GCV}(\lambda) = n^{-1} \sum_i (y_i - \hat{f}(x_i))^2 / (1 - \text{df}/n)^2$.

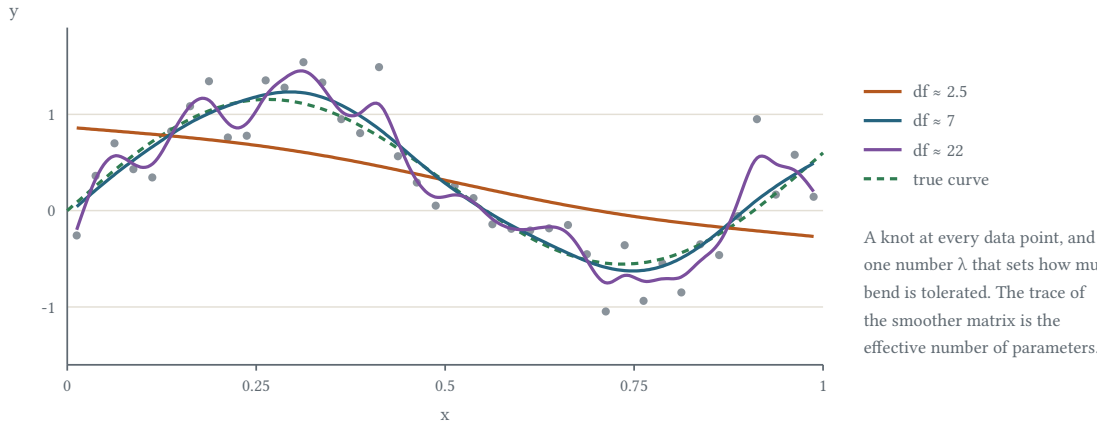


Figure 14: Smoothing splines fitted to the data of the previous figure at three effective degrees of freedom: nearly a straight line, close to the truth, and following the noise. The penalty λ moves the fit along this range.

11.3 Additive models and backfitting

DEFINITION Additive model $E[Y | \mathbf{x}] = \alpha + \sum_{j=1}^p f_j(x_j)$ with $E f_j(X_j) = 0$ for identifiability. The generalized additive model applies a link: $g(\mu) = \alpha + \sum_j f_j(x_j)$.

The backfitting algorithm sets $\hat{\alpha} = \bar{y}$, $f_j \equiv 0$, and cycles over j : $f_j \leftarrow S_j(\mathbf{y} - \hat{\alpha} - \sum_{k \neq j} f_k)$, then centres f_j , until the functions stop changing. For symmetric smoothers with eigenvalues in $[0, 1]$, such as smoothing splines, it converges, and for smoothing splines the limit minimizes $\sum_i (y_i - \alpha - \sum_j f_j(x_{ij}))^2 + \sum_j \lambda_j \int f_j''^2$ (Buja, Hastie and Tibshirani, 1989); the solution is unique unless a function of some predictors is exactly an additive function of the others (concurvity). Proof omitted.

11.4 MARS

DEFINITION Hinge functions and MARS A hinge pair at knot t is $(x - t)_+$ and $(t - x)_+$, the positive parts. MARS fits $f(x) = \beta_0 + \sum_m \beta_m B_m(x)$ where each B_m is a product of hinge functions of distinct variables; the degree is the number of factors allowed.

The forward pass starts from $B_0 = 1$ and, at each step, adds the pair $B_l(x_v - t)_+, B_l(t - x_v)_+$ over all existing terms B_l , variables v and observed values t , choosing the one that lowers RSS most, until a maximum number of terms is reached. The backward pass removes one term at a time, each time the one whose removal raises RSS least, and the model size is chosen to minimize $\text{GCV} = (\text{RSS} / N) / (1 - C(M) / N)^2$ with $C(M) = M + dK$, K the number of knots retained and d a fixed cost per knot, typically between 2 and 4. With degree 1 the model is additive with piecewise linear components.

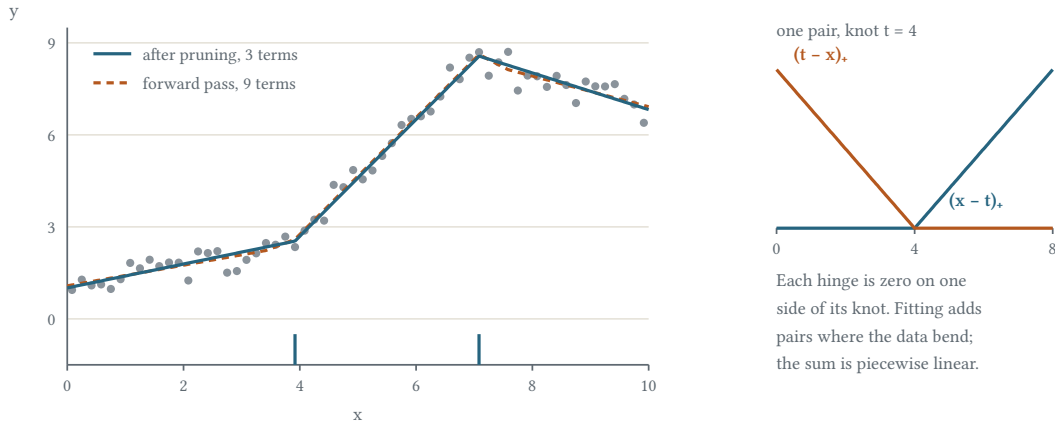


Figure 15: MARS in one dimension. Right: the hinge pair at knot $t = 4$. Left: the forward pass overgrows a sum of hinges (dashed, nine terms) and the backward pass prunes it to three, with knots near the two bends of the data.

12 Robust statistics

In the notes: Robust Statistics

12.1 Breakdown and the influence function

DEFINITION Breakdown point The finite-sample breakdown point of an estimator T at a sample is the smallest fraction m/n of observations that can be replaced by arbitrary values so that T becomes unbounded.

The mean has breakdown $1/n$: moving one point to infinity moves the mean. The median has breakdown $\lceil n/2 \rceil / n \rightarrow 1/2$: fewer than half of the points cannot push both order statistics around the middle beyond the remaining ones.

DEFINITION Influence function For a functional T and distribution F , $\text{IF}(x; T, F) = \lim_{\varepsilon \rightarrow 0} [T((1 - \varepsilon)F + \varepsilon\delta_x) - T(F)]/\varepsilon$, the effect on T of a small contamination at x . For regular T , $\sqrt{n}(T(F_n) - T(F)) \xrightarrow{d} \mathcal{N}(0, E[\text{IF}^2])$ (Hampel et al., 1986), and $\sup_x |\text{IF}|$ is the gross-error sensitivity.

EXAMPLE Mean and median Mean: $T(F_\varepsilon) = (1 - \varepsilon)\mu + \varepsilon x$, so $\text{IF} = x - \mu$, unbounded. Median m : $F_\varepsilon(m_\varepsilon) = 1/2$ means $(1 - \varepsilon)F(m_\varepsilon) + \varepsilon \mathbb{1}\{x \leq m_\varepsilon\} = 1/2$. Differentiating at $\varepsilon = 0$, with $F(m) = 1/2$, gives $-1/2 + f(m)m' + \mathbb{1}\{x \leq m\} = 0$, so $\text{IF} = \text{sign}(x - m)/(2f(m))$, bounded.

12.2 M-estimators

DEFINITION M-estimator of location $\hat{\theta} = \arg \min_{\theta} \sum_i \rho(x_i - \theta)$, equivalently the root of $\sum_i \psi(x_i - \theta) = 0$ with $\psi = \rho'$. Scale is estimated separately, for example by $\text{MAD} / 0.6745$.

THEOREM Influence of an M-estimator At F with $E_F \psi(X - \theta_0) = 0$, $\text{IF}(x) = \psi(x - \theta_0) / E_F \psi'(X - \theta_0)$, and the asymptotic variance is $E \psi^2 / (E \psi')^2$.

Proof. θ_ε solves $g(\varepsilon, \theta) = (1 - \varepsilon) E_F \psi(X - \theta) + \varepsilon \psi(x - \theta) = 0$. At $(0, \theta_0)$, $\partial_\varepsilon g = \psi(x - \theta_0)$ and $\partial_\theta g = -E_F \psi'(X - \theta_0)$; implicit differentiation gives $\theta' = -\partial_\varepsilon g / \partial_\theta g$. The variance is $E[\text{IF}^2]$. $\square$

ESTIMATOR	$\Psi(U)$	INFLUENCE
Mean	u	unbounded
Median	$\text{sign}(u)$	bounded, jumps
Huber, k	$\max(-k, \min(k, u))$	bounded; $k = 1.345$ gives 95% efficiency at the normal
Tukey biweight, c	$u(1 - (u/c)^2)^2$ for $ u \leq c$, else 0	redescends to 0; $c = 4.685$ for 95% efficiency

Huber's ψ is monotone, so ρ is convex and the solution is unique. The biweight is redescending: ρ is not convex, there can be several roots, and a robust starting value is required. Iteratively reweighted least squares solves $\sum_i w_i(x_i - \theta) = 0$ with weights $w_i = \psi(r_i/s)/(r_i/s)$, refitted until convergence; for Huber's ρ every step decreases the objective.

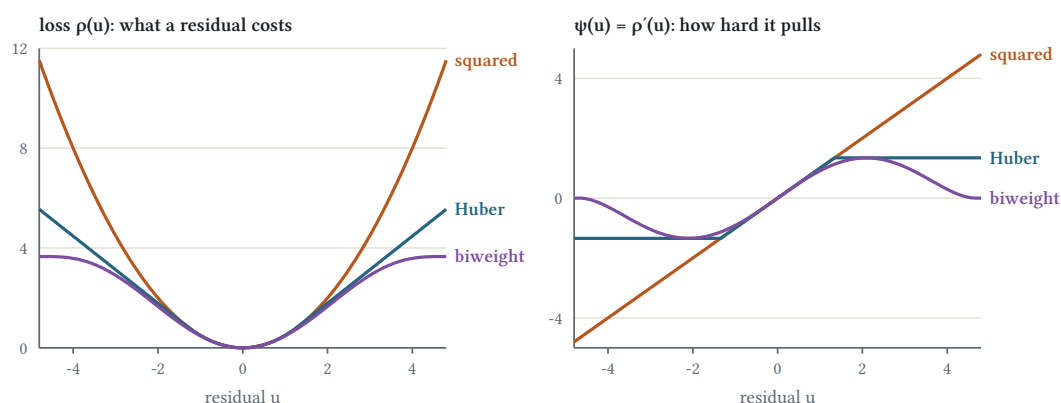


Figure 16: Loss ρ (left) and its derivative ψ (right), the pull of a residual on the fit. Squared error pulls proportionally to the residual; Huber caps the pull; the biweight lets it fall back to zero for gross outliers.

12.3 Robust regression

An M-estimator of regression minimizes $\sum_i \rho(r_i(\beta)/s)$. Its influence function is proportional to $\psi(r/\sigma)x$: bounded in the residual but not in x . A single point with extreme leverage can therefore still break it down, with breakdown point $1/n$. Bounded-influence (Mallows and Schweppe) estimators downweight by leverage as well. High-breakdown estimators are the least trimmed squares (LTS), which minimizes the sum of the h smallest squared residuals (breakdown near $1/2$ for $h \approx (n + p + 1)/2$), and MM-estimators, which start from a high-breakdown scale and fit and then take an efficient M-step (breakdown $1/2$, 95% efficiency at the normal); see Rousseeuw and Leroy (1987) and Maronna, Martin and Yohai (2019).

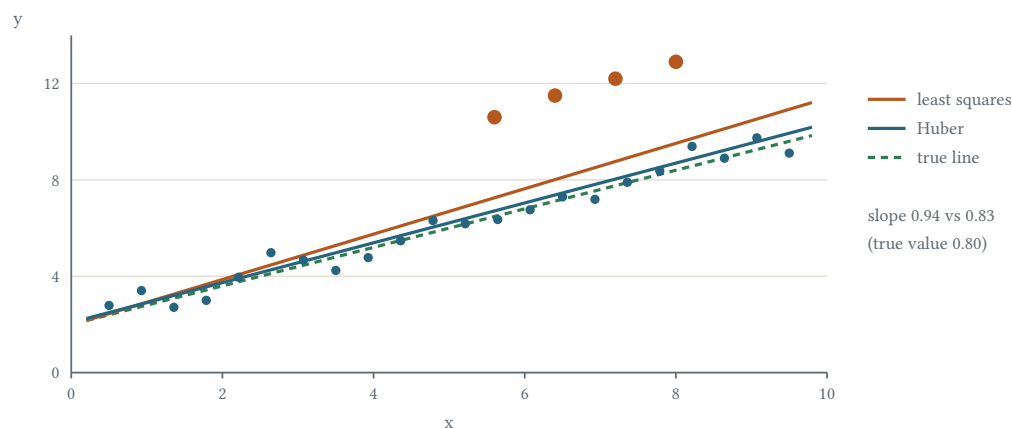


Figure 17: Four outliers in y . The least squares line is drawn towards them; the Huber M-estimate stays with the bulk of the data and recovers the true slope almost exactly.

13 Summary of the main formulas

QUANTITY	RESULT	WHERE EXPLAINED
Standard error of the mean	$\sigma/\sqrt{n}$	Sampling distributions
S^2 unbiased	$E S^2 = \sigma^2$, divisor $n - 1$	Point estimation
Normal sampling laws	$(n - 1)S^2/\sigma^2 \sim \chi_{n-1}^2$, $T \sim t_{n-1}$, $\bar{X} \perp S^2$	Sampling distributions
MSE	$\text{Var} + \text{bias}^2$	Point estimation
Cramér–Rao	$\text{Var}(\hat{\theta}) \geq 1/(nI(\theta))$	Point estimation
MLE	$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, I(\theta_0)^{-1})$	Point estimation
CI for the mean	$\bar{X} \pm t_{n-1, 1-\alpha/2} S/\sqrt{n}$	Confidence intervals
CI for the variance	$[(n - 1)S^2/\chi_{n-1, 1-\alpha/2}^2, (n - 1)S^2/\chi_{n-1, \alpha/2}^2]$	Confidence intervals
Wald interval	$\hat{\theta} \pm z_{1-\alpha/2} \hat{\text{se}}$	Confidence intervals
Sample size, one-sided z-test	$n = ((z_{1-\alpha} + z_{1-\beta})\sigma/\delta)^2$	Hypothesis tests
Likelihood ratio	$-2 \log \Lambda \xrightarrow{d} \chi_r^2$	Hypothesis tests
Bonferroni / BH	$p_i \leq \alpha/m$ / largest k with $p_{(k)} \leq k\alpha/m$	Hypothesis tests
Least squares	$\hat{\beta} = (X^\top X)^{-1} X^\top \mathbf{y}$, $\text{Var}(\hat{\beta}) = \sigma^2 (X^\top X)^{-1}$, $s^2 = \text{RSS} / (n - p)$	Linear regression
Coefficient t and interval	$(\hat{\beta}_j - \beta_j)/\hat{\text{se}}(\hat{\beta}_j) \sim t_{n-p}$	Linear regression
R^2 and overall F	SSR / SST ; $F = (\text{SSR} / (p - 1)) / (\text{SSE} / (n - p))$	Linear regression
Leverage	$h_{ii} = \mathbf{x}_i^\top (X^\top X)^{-1} \mathbf{x}_i$, $\sum_i h_{ii} = p$	Diagnostics
Cook's distance	$D_i = r_i^2 h_{ii} / (p(1 - h_{ii}))$	Diagnostics
Variance inflation	$\text{VIF}_j = 1/(1 - R_j^2)$	Diagnostics
Ridge	$\hat{\beta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top \mathbf{y}$, $\text{df} = \sum_j d_j^2 / (d_j^2 + \lambda)$	Regularization
Lasso, orthonormal	$\text{sign}(b_j)(b_j - \lambda)_+$	Regularization
Smoothing spline	$\hat{\mathbf{f}} = (I + \lambda K)^{-1} \mathbf{y}$, $\text{df} = \text{tr } S_\lambda$	Splines
Cubic spline dimension	$K + 4$ (natural: K)	Splines
Influence of an M-estimator	$\psi(x - \theta) / E \psi'$	Robust statistics

14 References

- G. Casella and R. L. Berger, *Statistical Inference*, 2nd ed., Duxbury, 2002: the standard source for the probability and distribution theory (chapters 1 to 5) and for the proofs of the Cramér–Rao bound, the MLE asymptotics and the Neyman–Pearson lemma.
- J. L. Devore, *Probability and Statistics for Engineering and the Sciences*, Cengage, 2011: the course text for the first part of the syllabus (chapters 1 to 6).
- L. Wasserman, *All of Statistics*, Springer, 2004: concise statements of the limit theorems, the bootstrap, MLE and multiple testing.
- B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*, Chapman and Hall, 1993.
- Y. Benjamini and Y. Hochberg, “Controlling the false discovery rate”, *J. R. Statist. Soc. B* 57 (1995).

- G. A. F. Seber and A. J. Lee, *Linear Regression Analysis*, 2nd ed., Wiley, 2003: the projection treatment of least squares, the F -test and the diagnostics.
- T. Hastie, R. Tibshirani and J. Friedman, *The Elements of Statistical Learning*, 2nd ed., Springer, 2009: ridge, lasso, splines, additive models and MARS (chapters 3, 5, 9).
- A. E. Hoerl and R. W. Kennard, "Ridge regression: biased estimation for nonorthogonal problems", *Technometrics* 12 (1970); R. Tibshirani, "Regression shrinkage and selection via the lasso", *J. R. Statist. Soc. B* 58 (1996).
- P. J. Green and B. W. Silverman, *Nonparametric Regression and Generalized Linear Models*, Chapman and Hall, 1994: smoothing splines and the Reinsch algorithm.
- J. H. Friedman, "Multivariate adaptive regression splines", *Annals of Statistics* 19 (1991); A. Buja, T. Hastie and R. Tibshirani, "Linear smoothers and additive models", *Annals of Statistics* 17 (1989).
- F. R. Hampel et al., *Robust Statistics: The Approach Based on Influence Functions*, Wiley, 1986; R. A. Maronna, R. D. Martin and V. J. Yohai, *Robust Statistics: Theory and Methods*, 2nd ed., Wiley, 2019; P. J. Rousseeuw and A. M. Leroy, *Robust Regression and Outlier Detection*, Wiley, 1987.